



**COMSATS University Islamabad,  
Sahiwal Campus.**

**SAMS**

**A Project Presented To**

**COMSATS University Islamabad,  
Sahiwal Campus**

**In partial fulfillment**

**of the requirement for the degree of**

**Bachelor of Science in Computer Science (2022-2026)**

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**The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of other**

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# CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS(CS) “**SAMS**” was developed by Faiza Sher (**CIIT/SP22-BCS-151/SWL**), **Samia Fatima** (**CIIT/SP22-BCS-177/SWL**) under the supervision of “**Ms. Samra Firdous Sabir**” and co supervisor “**Dr. Fakhar Mustafa**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Science.

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# Executive Summary

The Student Affairs Management System (SAMS) is a critical software project designed to digitize, streamline, and centralize the management of non-academic student support services at COMSATS University Islamabad. The primary objective of SAMS is to replace manual, paper-based processes with an efficient, reliable, and transparent digital platform.

The scope of SAMS is focused on delivering three core functional areas: Grievance Management, Event Management, and Counseling Services. ultimately aiming to enhance student engagement, improve administrative efficiency, and foster a supportive campus environment.

The design and implementation of SAMS follow an Agile Methodology using an Iterative Model, which emphasizes continuous feedback and incremental development to ensure the final product aligns closely with user needs.

The user-friendly interface ensures that students of all technical backgrounds can easily navigate the application. The successful implementation of SAMS will serve as a foundational tool for the Student Affairs Department to effectively monitor and respond to the diverse needs of the student body.

The SAMS system ensures strong security by encrypting student data, following data protection rules. It will be portable so it works smoothly on different devices, operating systems, and modern web browsers. SAMS reliability and high-performance, will give accurate results, staying available without errors.

In summary, the SAMS project represents a strategic investment in student welfare and administrative modernization at the university. By integrating essential student support functions into a secure, reliable, and user-centric digital environment.

# Acknowledgment

All praise is to almighty Allah who bestowed upon us a minute portion of his boundless knowledge by which we were able to accomplish this challenging task.

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And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

# Abbreviations

|              |                                         |
|--------------|-----------------------------------------|
| <b>CUI</b>   | Comsats University Islamabad            |
| <b>SRS</b>   | Software Require Specification          |
| <b>SDD</b>   | Software Design Description             |
| <b>UC</b>    | Use Case                                |
| <b>HCI</b>   | Human-Computer Interaction              |
| <b>DBMS</b>  | Database Management System              |
| <b>UT</b>    | Unit Testing                            |
| <b>FT</b>    | Functional Testing                      |
| <b>IT</b>    | Integration Testing                     |
| <b>NoSQL</b> | Not Only SQL (Flexible Database Format) |
| <b>IDE</b>   | Integrated Development Environment      |
| <b>CRUD</b>  | Create, Read, Update, Delete            |

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# **Chapter 1**

## **Introduction**

# **1 Introduction**

SAMS is a transformative digital platform designed to address the challenges faced by university students throughout their academic journey. Built specifically for COMSATS University Islamabad (CUI), Sahiwal Campus, SAMS provides a centralized space where students can manage grievances, register for events, and request counseling services all within a secure and user-friendly environment.

Unlike generic university portals, SAMS is uniquely structured to meet the needs of student affairs management. It enables students to submit grievances with proper documentation, browse and register for official university events, and schedule appointments for academic, personal, or career counseling. This ensures that all interactions are meaningful, authenticated, and relevant to campus life.

On the administrative side, SAMS serves as an essential tool for university departments responsible for student welfare. Administrators often face challenges in handling grievances, managing event registrations, and assigning counselors manually. SAMS resolves these inefficiencies by providing automated workflows, role-based dashboards, and real-time notifications—ensuring that information reaches the right stakeholders promptly and transparently.

Currently, COMSATS University Islamabad (CUI), Sahiwal Campus does not have a unified platform that integrates grievance handling, event participation, and counseling services. This lack of digital connectivity often results in delays, miscommunication, and difficulty keeping students informed. SAMS addresses this gap by offering a comprehensive, structured, and easy-to-use system that enhances student engagement and operational efficiency.

## **1.1 Brief Overview**

The SAMS project aims to develop a web-based application exclusively for the students and administrators of COMSATS University Islamabad, Sahiwal Campus, enhancing how student affairs are managed across the university. SAMS is built using modern web technologies such as React.js, Node.js/Express.js, and MongoDB, ensuring security, scalability, and real-time

interaction. The project follows the Agile methodology, enabling iterative development and continuous improvement.

SAMS is a focused and comprehensive platform that enables students to submit grievances, stay updated about academic and extracurricular events, and request counseling sessions. It eliminates the challenges previously faced due to manual processes, delays, and fragmented communication channels. Through real-time notifications, centralized data management, and role-based authentication, SAMS ensures that students never miss important updates related to university affairs.

The platform stands out because it is not a generic student management system but a specialized solution for handling grievance management, event management, and counseling workflows. SAMS provides students with a convenient way to interact with administrative departments, ultimately improving institutional transparency and responsiveness.

## **1.2 Relevance to Course Modules**

The development of SAMS integrates various concepts and skills gained throughout the Computer Science degree program.

### **i. Technical Skills:**

SAMS involves applying programming skills and tools learned during the course, including JavaScript, React.js, Node.js/Express.js, and MongoDB. These technologies were essential for implementing core functionalities.

### **ii. Software Engineering Principles:**

The project adheres to core software engineering concepts including requirement analysis, SRS documentation, system design, Agile methodology, quality assurance, and testing. SAMS applies principles from courses like Software Engineering, Web Engineering, Database Systems, and HCI to ensure scalability and maintainability.

### **iii. Database Design:**

Understanding of database systems was required to design the SAMS database. Collections such as students, grievances, events, counselors, and notifications were designed with proper indexing and normalization to ensure efficiency and accuracy.

### **iv. System Architecture:**

The SAMS system uses a client-server architecture with RESTful APIs. Concepts like micro services, modular backend development, API communication, and cloud-hosted deployment were applied to create a structured and scalable system.

**v. UI/UX Design:**

Creating a simple and intuitive interface was essential for user adoption. UI/UX principles learned in Human-Computer Interaction and Interface Design courses helped create an ergonomic and user-friendly design for dashboards, forms, and notifications.

**vi. Project Management:**

Project management principles—planning, task division, sprint management, and progress tracking guided the structured development of SAMS. Tools and techniques learned in this course ensured timely delivery of each phase.

**vii. Networking:**

SAMS relies on web communication protocols, real-time updates, and server-client requests to deliver notifications and update records instantly. Understanding networking fundamentals helped in handling secure communication and backend connectivity.

### **1.3 Project Background**

SAMS was developed to address ongoing challenges faced by students and administrative departments at COMSATS University Islamabad, Sahiwal Campus. Before SAMS, there was no centralized digital system for submitting student grievances, registering for university events, or requesting counseling services. Students often relied on manual processes and fragmented communication channels, leading to delays, confusion, and inefficiencies.

University departments responsible for grievance redressal, academic affairs, and counseling also depended on paper-based or manually managed methods. This made it difficult to track cases, assign committees, update students, or manage appointments effectively.

SAMS was developed to overcome these limitations by providing a unified digital solution that allows students to:

Submit Grievance with detail and evidence, tracking grievance status with unique case ID, register for official university events, view an event calendar with filters, request for sessions and receive Notification for updates.



## 1.4 Literature Review

Table 1.1: Literature Review

| Application Name                 | Weakness                                                                                                                                                      | Proposed Project Solution (SAMS)                                                                                                                                      |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Campus Management Systems</b> | Mostly focused on academics, attendance, and administrative records. They do not integrate student affairs modules such as events, grievances, or counseling. | SAMS provides a centralized platform where student societies, event management, grievance handling, and counseling services are all integrated in one unified system. |
| <b>Event Booking Apps</b>        | Support only basic event registration; no workflow for society approvals, department verification, or event monitoring.                                       | SAMS includes event creation, multi-level approvals, event verification, and student participation tracking specifically designed for university environments.        |
| <b>Grievance Portals</b>         | Complex forms, slow response, lack of categorization, and no way for students to track the status of their complaints.                                        | SAMS offers simple grievance submission, automatic ticket generation, category-wise complaints, real-time status tracking, and a transparent resolution timeline.     |
| <b>Counseling Platforms</b>      | Not customized for university needs; lack proper appointment scheduling, privacy handling, and counselor-student communication flow.                          | SAMS provides a dedicated counseling module with appointment booking, counselor availability, session history, and secure communication.                              |

## **1.5 Analysis from Literature Review**

The review of existing campus, event, grievance, and counseling platforms reveals significant limitations in supporting student affairs holistically. Most existing systems are fragmented, focusing only on individual components such as academics, events, or complaint handling. SAMS (Student Affairs Management System) is designed to address these gaps by integrating all student-related services into a unified, smart, and user-friendly solution. Below is detailed analysis based on the identified weaknesses.

### **1.5.1 Campus Management System:**

Most Campus Management Systems primarily handle academic operations. SAMS bridges this limitation by introducing a centralized student affairs platform that integrates society management, event posting, approvals, grievance ticketing, and counseling functions that traditional lack. This allows universities to manage both academic and non-academic operations in one ecosystem.

### **1.5.2 Event Booking Applications**

Generic event registration apps support only basic sign-ups and notifications. They are not designed for university structures that require society-level requests, departmental approvals and verification steps.

SAMS enhances this process by introducing a multi-level event approval workflow, automated verification, event history tracking, and streamlined participation management tailored specifically to campus society events. This makes event handling more transparent, traceable, and efficient.

### **1.5.3 Grievance Portals**

Existing grievance portals are often slow, complex, and lack transparency. Students commonly face issues such as lengthy forms, unclear categories, and no way to track complaint progress.

SAMS addresses these issues by offering a simplified grievance submission system with categorized templates, automatic ticket generation, progress tracking, and clear resolution timelines. This ensures accountability and improves student trust in the grievance-handling process.

#### **1.5.4 Counseling Platforms**

Many standalone counseling platforms are too general and not adapted to university contexts. They lack proper privacy management, do not reflect counselor availability, and fail to create structured sessions.

SAMS overcomes these challenges by providing a dedicated counseling module with secure appointment scheduling, counselor availability calendars, private communication, and session logs. This results in better student support and improved mental health service management within the university.

### **1.6 Methodology and Software Lifecycle**

SAMS is a comprehensive web-based system with multiple dynamic modules such as grievance handling, event management, and counseling services. Because the project involves evolving requirements, multiple stakeholders (students, admins, counselors), and continuous improvements based on user feedback, the selected development methodology must support flexibility and iteration. For this reason, the Agile Software Development Methodology is the most appropriate choice for SAMS.

#### **1.6.1 Rationale behind Selected Agile Model**

Agile supports continuous improvement, regular feedback, and incremental development. For SAMS, involving university students, societies, and administrative staff throughout development is crucial to ensure the platform meets real user needs.

Agile principles applied to SAMS include:

**Iterative Development:** Each sprint adds new modules like events, grievances, or counseling, enabling early testing and improvement.

**Incremental Delivery:** Functional components are released gradually, allowing stakeholders to use and review them during development.

**Continuous Feedback:** Student and staff feedback guides each iteration, ensuring SAMS remains relevant and user-friendly.

**Adaptive Planning:** Requirements for student affairs are dynamic; Agile supports adjustments whenever policies or needs change.

## **1.7 Summary**

This chapter explains the project's purpose, its connection to course concepts, and insights from related research. It also describes the chosen method and why it fits the project.

# **Chapter 2**

## **Problem Definition**

## 2 Problem Definition

This chapter discusses the precise problem to be solved. It should extend to include the outcome.

### 2.1 Problem Statement

At COMSATS University Islamabad, Sahiwal Campus, students often face difficulties in accessing and managing essential student-related services due to the absence of a centralized digital platform. Currently, major services such as grievance submission, event registration, and counseling appointments are handled through manual processes, physical visits, informal communication channels, and scattered record-keeping. This leads to delays, miscommunication, low student engagement, and inefficiency in managing student affairs.

Students have no structured or reliable way to submit grievances, track their status, or receive timely updates. This often results in unresolved complaints, lack of transparency, and student dissatisfaction. Event participation is also affected because students rely on notice boards, WhatsApp groups, and verbal communication to learn about upcoming university events—causing many students to miss important activities. Likewise, there is no streamlined system for requesting academic, personal, or career counseling sessions, which makes it difficult for students to seek support when needed.[1]

From the administrative perspective, handling grievances manually is time-consuming and prone to errors. Managing event registrations, assigning committees, updating students, and scheduling counseling sessions becomes challenging without a unified system. Due to the lack of automation, administrators struggle to maintain accurate records, send notifications promptly, and ensure efficient service delivery.

To resolve these challenges, SAMS is being developed as a unified platform that centralizes and streamlines all student services

For administrators, SAMS ensures efficient management through automated grievance assignment and review, digital event creation and monitoring, role-based access for

counselors, centralized dashboards for decision-making and Real-time communication with students

The absence of a dedicated and intelligent system highlights the need for SAMS. By offering a comprehensive digital platform, SAMS ensures transparency, improves institutional efficiency, enhances student engagement, and supports a more organized and responsive student affairs ecosystem.

## **2.2 Deliverables and Development Requirements**

The SAMS project includes a comprehensive set of development requirements and deliverables to ensure the successful implementation of a secure, efficient, and user-friendly student affairs platform. These deliverables cover UI design, system functionality, backend services, documentation, and long-term maintenance to support continuous improvement of the system.

### **2.2.1 User Interface (UI) Design**

Design an intuitive, responsive, and accessible web interface that enables students, administrators, and counselors to navigate the system effortlessly. The UI must offer a clean layout for submitting grievances, browsing events, and requesting counseling sessions while ensuring a visually appealing and user-friendly experience.

### **2.2.2 User Registration & Authentication**

Implement a secure user authentication system using role-based access control. Students, admins, and counselors should be able to log in with verified credentials. Admins and counselors receive elevated permissions, while students access only the features relevant to them.

### **2.2.3 Grievance Management Module**

Provide a structured interface where students can submit grievances with details and attachments. Admins should be able to view, filter, approve/reject grievances, assign

committees, and manage grievance statuses. Students can track their grievance using a unique case ID.

#### **2.2.4 Event Management Module**

Develop a centralized system where administrators can create, update, and manage university events. Students can browse events through an interactive event calendar and register for events, ensuring timely access to all upcoming university activities. The system will support multi-level event approval workflows, allowing societies to submit event proposals that are reviewed by department heads and administrators. Real-time notifications will keep students updated about event changes, deadlines, and newly added activities. This centralized approach improves transparency, enhances student engagement, and ensures efficient coordination between societies and university management.

#### **2.2.5 Counseling Management Module**

Enable students to request academic, personal, or career counseling sessions. Admins can assign counselors, and counselors can conduct sessions, update records, and view feedback. Students receive notifications regarding scheduled appointments.[2]

#### **2.2.6 Notifications System**

Implement automated real-time notifications for grievance updates, event reminders, and counseling appointments. Notifications are delivered through the system dashboard and stored for future reference.

#### **2.2.7 Search and Filtering Features**

Develop efficient search and filtering mechanisms, allowing admins to filter grievances based on status, date, or category, and allowing students to search events or counseling options easily.



### **2.2.8 Security and Privacy**

Prioritize data security through encrypted data storage, secure login mechanisms, and strict role-based access controls. SAMS must comply with data protection standards to safeguard student information and ensure confidentiality.

### **2.2.9 Scalability and Performance Optimization**

Ensure the platform supports increasing numbers of grievances, events, students, and counseling requests. The system must be optimized for fast loading, high uptime, and scalable backend operations through technologies such as Node.js and MongoDB.

### **2.2.10 Web Application Development**

As SAMS is a web-based platform, it will be developed using React.js for the frontend and Node.js with Express.js for the backend. MongoDB will be used for database management, ensuring real-time updates and seamless data flow across the system.

SAMS will include clear onboarding tutorials that guide users through each module step-by-step. Interactive walkthroughs and tooltips will be integrated within the interface to help users learn features quickly. Regularly updated FAQs and support articles will ensure that students, admins, and counselors can resolve common issues without external assistance.

### **2.2.11 Documentation**

Prepare detailed documentation including SRS, SDD, user manuals, and developer documentation to support future development, maintenance, and onboarding of new developers.

### **2.2.12 Testing and Quality Assurance**

Conduct comprehensive testing procedures such as unit testing, integration testing, usability testing, and user acceptance testing. This ensures that all modules: grievances, events, and counseling function correctly and meet user expectations.

### **2.2.13 User Training and Support Materials**

Provide training resources such as user guides, tutorials, FAQs, and assistive documentation to help students, admins, and counselors understand and efficiently use the SAMS platform.

### **2.2.14 Maintenance, Updates, and Continuous Improvement**

Develop a long-term maintenance plan that includes fixing bugs, updating modules, implementing new features based on student and admin feedback, enhancing security, and ensuring smooth operation of the system.

## **2.3 Current System**

A centralized system is currently missing at COMSATS Sahiwal, making it difficult for students to submit grievances, register for events, or request counseling services in an organized way. Because these processes depend on manual methods and informal communication channels, information often becomes delayed, disorganized, or unclear. Students also have no structured way to track grievance progress, view official university events, or schedule counseling sessions efficiently. To overcome these challenges, SAMS is required.

# **Chapter 3**

## **Requirement Analysis**

## 3 Requirement Analysis

### 3.1 Use Case Diagrams

This use case diagram depicts Grievance Management, Event Management, and Student Counseling, highlighting user interactions and key functionalities.

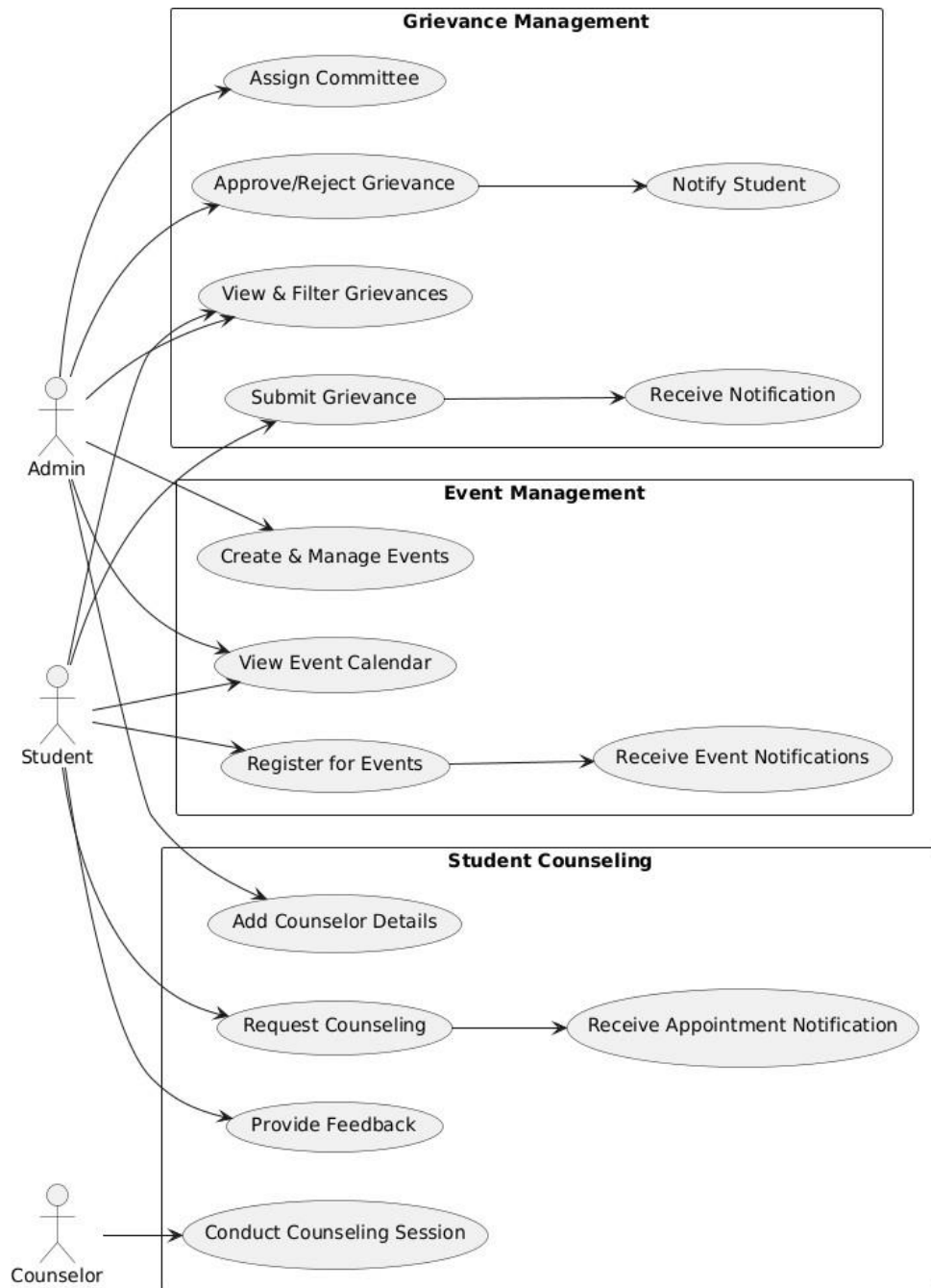


Figure 3.1: Use Case Diagram

## 3.2 Use Case Descriptions

The table below indicate comprehensive use case descriptions

The table below allows the admin to assign members for grievance handling after logging in.

The system updates records, notifies members, and ensures only authorized admins can perform this action, with error handling for unauthorized access.

**Table 3.1: Assign Committee**

|                  |                                                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 01                                                                                                                                                                                                                                           |
| Use case name    | Assign Committee                                                                                                                                                                                                                             |
| Actors           | Admin                                                                                                                                                                                                                                        |
| Description      | The admin assigns a committee responsible for handling student grievances.                                                                                                                                                                   |
| Trigger          | Admin initiates the committee assignment.                                                                                                                                                                                                    |
| Pre-condition    | Admin must be logged in                                                                                                                                                                                                                      |
| post condition   | Committee members are assigned, and notifications are sent.                                                                                                                                                                                  |
| Normal flow      | Admin logs into the system.<br><br>Admin navigates to the "Assign Committee" section.<br>Admin selects members for the committee.<br>Admin confirms the assignment.<br><br>The system updates the records and notifies the assigned members. |
| Alternative flow | Admin can modify selections before confirming.                                                                                                                                                                                               |
| Exceptions       | If the admin is unauthorized, access is denied.<br><br>If the system fails to save, an error message appears.                                                                                                                                |
| Business rules   | Only authorized admins can assign a committee.<br><br>Each grievance must have an assigned committee.                                                                                                                                        |

The Approve/Reject Grievance use case allows the admin to review, approve, or reject student grievances. The system updates the status, notifies the student, and enforces rules like restricting unauthorized access and preventing resubmission of rejected grievances.

**Table 3.2: Approve/Reject Grievance**

|                  |                                                                                                                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 02                                                                                                                                                                                                                     |
| Use case name    | Approve/Reject Grievance                                                                                                                                                                                               |
| Actors           | Admin                                                                                                                                                                                                                  |
| Description      | Admin reviews and updates the status of student grievances.                                                                                                                                                            |
| Trigger          | Admin selects a grievance for review.                                                                                                                                                                                  |
| Pre-condition    | A grievance must be submitted.                                                                                                                                                                                         |
| post condition   | The grievance status is updated, and the student is notified.                                                                                                                                                          |
| Normal flow      | Admin logs in and navigates to the grievance section.<br>Admin selects a grievance to review.<br>Admin approves or rejects it.<br><br>The system updates the grievance status.<br>The student receives a notification. |
| Alternative flow | The admin can leave the grievance as "Pending" for further review.                                                                                                                                                     |
| Exceptions       | Unauthorized access prevents grievance approval.<br>If grievance ID is invalid, an error is displayed.                                                                                                                 |
| Business rules   | Only the admin can approve or reject grievances.<br>Once a grievance is rejected, it cannot be resubmitted.                                                                                                            |

The table View & Filter Grievances enables the admin to access, filter, and review student grievances based on various criteria. The system displays relevant grievances, allowing the admin to take necessary actions while handling errors like no matching results or retrieval failures.

The View & Filter Grievances feature allows administrators to search and organize grievances using filters such as category, date, priority, or status. This helps admins quickly locate specific complaints and prioritize those that require urgent attention. The interface presents each grievance with complete details, including student information, submission time, and issue description. Additionally, the system displays warning messages when no records match the selected filters and handles retrieval errors gracefully to ensure smooth operations. This ensures efficient grievance monitoring and faster resolution across all departments.

Integration with the notification system ensures that students are informed about the status and resolution of their complaints.

**Table 3.3: View & Filter Grievances**

|                  |                                                                                                                                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 03                                                                                                                                                                                                                                                                                                                    |
| Use case name    | View & Filter Grievances                                                                                                                                                                                                                                                                                              |
| Actors           | Admin                                                                                                                                                                                                                                                                                                                 |
| Description      | Admin views and filters student grievances using different criteria.                                                                                                                                                                                                                                                  |
| Trigger          | The admin accesses the grievance management system.                                                                                                                                                                                                                                                                   |
| Pre-condition    | The admin must be logged in.<br>There must be grievances submitted by students.                                                                                                                                                                                                                                       |
| Post condition   | The grievances are displayed based on the selected filters.<br>The admin can take necessary actions based on the grievance details.                                                                                                                                                                                   |
| Normal flow      | Admin logs into the system.<br>Admin navigates to the "Grievance Management" section. System displays a list of all grievances.<br>Admin selects filtering criteria (e.g., status, date, category). The system filters and displays relevant grievances.<br>Admin reviews the grievances and takes necessary actions. |
| Alternative flow | If the admin modifies filters, the system updates the grievance list.                                                                                                                                                                                                                                                 |
| Exceptions       | If no grievances match the selected filter, the system displays: "No grievances found."<br>If the system fails to retrieve grievances, an error message appears.                                                                                                                                                      |
| Business rules   | Only authorized admins can access grievances.<br>Grievances should be categorized and searchable based on multiple filters                                                                                                                                                                                            |

Notify Student use case ensures that students receive timely notifications about grievance updates. The system automatically detects status changes, sends notifications, queues them if the student is offline, and logs errors if delivery fails.

The Notify Student use case ensures that students stay informed about any updates or actions taken on their submitted grievances. Notifications are sent instantly when the status of grievance changes, such as when it is reviewed, resolved, or requires additional information.

If a student is offline, the system queues the notification and delivers it once they log in. All notification attempts and failures are logged for auditing and troubleshooting purposes. This feature enhances transparency and keeps students engaged in the grievance process.

**Table 3.4: Notify Students**

|                  |                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 04                                                                                                                                           |
| Use case name    | Notify Student                                                                                                                               |
| Actors           | System                                                                                                                                       |
| Description      | The system notifies the student about grievance updates.                                                                                     |
| Trigger          | Change in grievance status.                                                                                                                  |
| Pre-condition    | The grievance must be processed.                                                                                                             |
| Post condition   | Student receives a notification.                                                                                                             |
| Normal flow      | The system detects an updated grievance status.<br>The system generates a notification.<br>The system sends the notification to the student. |
| Alternative flow | If the student is offline, the notification is queued.                                                                                       |
| Exceptions       | If notification sending fails, an error is logged.                                                                                           |
| Business rules   | Notifications must be sent within 24 hours of grievance update.                                                                              |

This use case allows students to log in, enter grievance details, and submit them for review. The system records the grievance, notifies the admin, supports drafts, enforces required fields, and restricts students to one grievance submission at a time.

This use case enables students to securely log in and submit grievances through a structured form, ensuring all required details are captured accurately. The system validates mandatory fields and prevents incomplete submissions, while also allowing students to save drafts for later editing. Once submitted, the grievance is automatically recorded in the database and the relevant admin is notified for timely review. Students are restricted to one active grievance at a time to avoid duplication and confusion. The system tracks submission timestamps, updates the status of each grievance in real time, and provides students with a view of their grievance history. Additionally, the interface offers guidance and tooltips to help students describe their issues clearly, improving the efficiency and accuracy of the grievance handling process.



Audit logs maintain records of all actions for accountability, and error handling ensures that any submission or system failure is reported and resolved efficiently. This comprehensive approach improves grievance accuracy, speeds up resolution, and enhances student confidence in the system. The grievance is automatically stored in the database and the relevant administrator is notified for prompt review.

**Table 3.5: Submit Grievance**

|                  |                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 05                                                                                                                                                                                       |
| Use case name    | Submit Grievance                                                                                                                                                                         |
| Actors           | Student                                                                                                                                                                                  |
| Description      | A student submits a grievance regarding an issue                                                                                                                                         |
| Trigger          | User requests to log in or register.                                                                                                                                                     |
| Pre-condition    | The student initiates a grievance submission.                                                                                                                                            |
| Post condition   | The student must be logged in.                                                                                                                                                           |
| Normal flow      | <p>Student logs in.</p> <p>Student selects "Submit Grievance."</p> <p>Student enters grievance details and submits.</p> <p>System records the grievance and notifies Admin.</p> <p>.</p> |
| Alternative flow | Student can save as draft before final submission.                                                                                                                                       |
| Exceptions       | If required fields are missing, an error appears..                                                                                                                                       |
| Business rules   | Students can submit only one grievance at a time.                                                                                                                                        |

The Receive Notification use case ensures that students get a confirmation after submitting a grievance. The system verifies the submission, sends a notification within 5 minutes, retries if it fails, and logs errors for missing contact details or notification failures.

The Receive Notification use case ensures that students receive a timely confirmation after submitting a grievance, providing assurance that their issue has been recorded. The system verifies the submission details, generates a confirmation notification within five minutes, and retries automatically if delivery fails. Any errors, such as missing contact information or failed notification attempts, are logged for administrative review. Notifications are displayed both within the SAMS dashboard and, if enabled, via email to ensure the student does not

miss updates. This feature enhances transparency and keeps students informed from the moment of submission, reducing uncertainty about the grievance process.

The system also maintains a history of all notifications sent for each grievance, allowing students to verify that confirmations were received. Students can view the notification details directly in their SAMS account, including timestamp and status. The system ensures that notifications are concise, clear, and consistent with university policies.[3]

**Table 3.6: Receive Notification**

|                  |                                                                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 06                                                                                                                                                             |
| Use case name    | Receive Notification                                                                                                                                           |
| Actors           | Student                                                                                                                                                        |
| Description      | The system sends a notification to the student after submitting a grievance..                                                                                  |
| Trigger          | A student submits a grievance.                                                                                                                                 |
| Pre-condition    | The student must be logged in.<br>The grievance must be successfully submitted.                                                                                |
| Post condition   | The student receives a confirmation notification<br>The grievance is recorded in the system.                                                                   |
| Normal flow      | Student submits a grievance. System verifies the submission<br>System sends a confirmation notification to the student. The student receives the notification. |
| Alternative flow | If the notification fails, the system retries sending it                                                                                                       |
| Exceptions       | If the system fails to send a notification, an error log is created.<br>If the student's contact information is missing, an error is displayed.                |
| Business rules   | The student must receive a notification within 5 minutes of submission.<br>Notifications must be stored in the system for future reference.                    |

The given use case allows the admin to create and manage student events. The system stores event details, enforces a 7-day advance scheduling rule, allows editing before saving, and displays errors for missing information. The given use case allows the admin to create, update, and manage student events efficiently within the SAMS platform. The system securely stores all event details, including title, description, date, time, venue, and organizing society. .

**Table 3.7: Create & Manage Events**

|                  |                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 07                                                                                                                                              |
| Use case name    | Create & Manage Events                                                                                                                          |
| Actors           | Admin                                                                                                                                           |
| Description      | The admin creates and manages student events.                                                                                                   |
| Trigger          | The admin initiates event creation.                                                                                                             |
| Pre-condition    | Admin must have event management permissions.                                                                                                   |
| Post condition   | Event details are stored in the system.                                                                                                         |
| Normal flow      | Admin logs in.<br>Admin selects "Create Event." Admin enters event details.<br>System saves the event. Students can now register for the event. |
| Alternative flow | Admin can edit event details before saving.                                                                                                     |
| Exceptions       | If the event details are missing, an error appears.                                                                                             |
| Business rules   | Events must be scheduled at least 7 days in advance.                                                                                            |

The Register for Events use case allows students to enroll in available events. The system confirms registration, permits cancellations before the event starts, enforces capacity limits, and closes registration 24 hours before the event. The Register for Events use case allows students to browse available events and enroll directly through the SAMS platform. Upon registration, the system immediately confirms the student's enrollment and updates the event's participant count.

Students are allowed to cancel their registration, but only before the event begin, ensuring accurate attendance tracking. The system enforces capacity limits, preventing overbooking and maintaining a safe and organized environment.

Registration automatically closes 24 hours prior to the event to give organizers time for final preparations. Students can view their registered events in a personalized dashboard, including event details such as date, time, and venue.

Notifications are sent for successful registration, cancellations, or if an event reaches full capacity. The system also maintains a history of registrations for auditing and future reference. Any errors, such as attempting to register for a full or past event, are displayed clearly to guide the student.

This feature streamlines student participation, improves event management efficiency, and ensures fairness and transparency in enrollment.

**Table 3.8: Register for Events**

|                  |                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 08                                                                                                                                    |
| Use case name    | Register for Events                                                                                                                   |
| Actors           | Student                                                                                                                               |
| Description      | A student registers for an event.                                                                                                     |
| Trigger          | The student selects an event for registration.                                                                                        |
| Pre-condition    | The event must be open for registration.                                                                                              |
| Post condition   | The student is enrolled in the event.                                                                                                 |
| Normal flow      | Student logs in.<br><br>Student browses available events.<br>Student selects an event and registers.<br>System confirms registration. |
| Alternative flow | Student can cancel registration before the event starts.                                                                              |
| Exceptions       | If event capacity is full, registration fails.                                                                                        |
| Business rules   | Registration closes 24 hours before the event.                                                                                        |

In the given use case of View Event Calendar students can access a calendar displaying upcoming events. The system retrieves and presents events in an organized format, allows filtering, and handles errors. If no events are available or fail to load, the system will display an error message.

The View Event Calendar use case allows students to access a centralized calendar displaying all upcoming university events. The calendar provides a clear visual overview of events organized by date, time, and society. Students can click on individual events to view detailed information, including the description, venue, and organizer contact. The system retrieves event data from the SAMS database and presents it in an organized and user-friendly format. Filters are available to help students search for events by category, department, or society. Students can also view events in daily, weekly, or monthly layouts to suit their planning preferences. The interface highlights registered events separately to help students track their participation.

**Table 3.9: View Event Calendar**

|                  |                                                                                                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 09                                                                                                                                                                                                                   |
| Use case name    | View Event Calendar                                                                                                                                                                                                  |
| Actors           | Student                                                                                                                                                                                                              |
| Description      | The student views a calendar displaying upcoming events.                                                                                                                                                             |
| Trigger          | The student selects the event calendar option.                                                                                                                                                                       |
| Pre-condition    | The student must be logged in.<br><br>Events must be available in the system.                                                                                                                                        |
| Post condition   | The student successfully views the calendar.<br><br>The system displays all upcoming events.                                                                                                                         |
| Normal flow      | Student logs into the system.<br><br>Student navigates to the "Event Calendar" section.<br><br>System retrieves and displays all upcoming events in a calendar format.<br><br>Student views event details if needed. |
| Alternative flow | If the student wants to filter events, the system updates the calendar accordingly.                                                                                                                                  |
| Exceptions       | If no events are available, the system displays a message: "No events scheduled."<br><br>If the system fails to load events, an error message appears.                                                               |
| Business rules   | Only registered students can access the event calendar.<br><br>Events must be displayed in an easy-to-read format.                                                                                                   |

In Receive Event Notification table, students get notified about upcoming events they registered for. The system sends notifications at least 24 hours before the event, logs them, skips disabled notifications, and handles errors if delivery fails or contact details are missing. The system ensures that repeated notifications are not sent unnecessarily, avoiding confusion or spam. Real-time updates allow students to be informed if an event is rescheduled,

canceled, or updated. This feature enhances student participation, ensures timely reminders, and improves overall event engagement within the university.

**Table 3.10: Receive Event Notification**

|                  |                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 10                                                                                                                                                                                                        |
| Use case name    | Receiving Event Notification                                                                                                                                                                              |
| Actors           | Student                                                                                                                                                                                                   |
| Description      | The student receives notifications about upcoming events they registered for.                                                                                                                             |
| Trigger          | The system detects an upcoming event.                                                                                                                                                                     |
| Pre-condition    | The student must be registered for the event.<br>Notifications must be enabled for the student.                                                                                                           |
| Post condition   | The student receives an event notification.<br>The notification is logged in the system.                                                                                                                  |
| Normal flow      | The event date approaches.<br>The system checks the registered students for that event.<br>The system sends notifications to the registered students.<br>The student receives and reads the notification. |
| Alternative flow | If a student has disabled notifications, no notification is sent.                                                                                                                                         |
| Exceptions       | If the system fails to send a notification, an error log is created.<br>If the student's contact information is missing, the system displays an error.                                                    |
| Business rules   | Notifications must be sent at least 24 hours before the event.<br>Students must be able to view past notifications in the system.                                                                         |

The Add Counselor detail use case allows the admin to add a counselor to a student requesting counseling. The system updates records, notifies the student, allows read counselor, and ensures each counselor handles a maximum of 10 students.

The system updates the student's records to reflect the assigned counselor, ensuring accurate tracking of all counseling requests. Once assigned, the student receives an automatic notification informing them of the counselor's details and appointment scheduling. The system allows students to view the assigned counselor's basic profile and contact information, while keeping sensitive details secure.

**Table 3.11: Add Counselor detail**

|                  |                                                           |
|------------------|-----------------------------------------------------------|
| Use case id      | 11                                                        |
| Use case name    | Add Counselor detail                                      |
| Actors           | Admin                                                     |
| Description      | The Admin adds counselor detail                           |
| Trigger          | A student requests counseling.                            |
| Pre-condition    | The student must have submitted a request.                |
| Post condition   | The counselor is added, and the student is notified.      |
| Normal flow      | Admin views counseling requests.<br>Admin adds counselor. |
| Alternative flow | Admin can read a different counselor                      |
| Exceptions       | If no counselors are available, admin cannot add detail.  |
| Business rules   | Each counselor can handle a maximum of 10 students.       |

Request Counseling allows students to submit a request for a counseling session. The system records the request, enforces one active request per student, allows edits before submission, and displays errors if required details are missing.

The Request Counseling use case enables students to submit a formal request for a counseling session through the SAMS platform. The system records each request in the database, ensuring that all necessary details such as student name, issue description, and preferred timing are captured accurately. To prevent multiple overlapping requests, the system enforces a rule allowing only one active counseling request per student at a time. Students can edit their request before final submission, ensuring that all information is complete and accurate. The system validates required fields and displays clear error messages if any details are missing or incomplete. Upon successful submission, an automatic notification is sent to both the student and the administration.

The system tracks the status of each request, allowing students to monitor whether it is pending, assigned, or completed. Historical data of past counseling requests is maintained for reference and auditing purposes.

Error handling ensures that system or network failures do not result in lost requests. This use case improves the efficiency of counseling services, ensures proper student support, and maintains transparency in the request and assignment process.

**Table 3.12: Request Counseling**

|                  |                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------|
| Use case id      | 12                                                                                                          |
| Use case name    | Request Counseling                                                                                          |
| Actors           | Student                                                                                                     |
| Description      | A student requests a counseling session.                                                                    |
| Trigger          | The student submits a counseling request.                                                                   |
| Pre-condition    | Student must be logged in.                                                                                  |
| Post condition   | The request is recorded in the system.                                                                      |
| Normal flow      | Student selects "Request Counseling."<br>Student provides details and submits.<br>System saves the request. |
| Alternative flow | Student can edit request before submission.                                                                 |
| Exceptions       | If required details are missing, an error appears.                                                          |
| Business rules   | Students can only request one session at a time.                                                            |

Provide Feedback use case allows students to submit feedback after a completed counseling session. The system stores feedback securely, notifies the counselor, allows edits before final submission, and restricts feedback to completed sessions only.

The Provide Feedback use case enables students to submit detailed feedback after completing a counseling session. The system ensures that feedback is stored securely in the database, protecting student privacy. Students can access their previous feedback and make edits before final submission if needed. The system validates the feedback to ensure all required fields are filled. Only sessions that have been completed can receive feedback, preventing incomplete



or invalid submissions. Once feedback is submitted, the counselor is automatically notified to review the comments. The system logs the submission time and maintains a record for future reference. Error messages are displayed if the feedback cannot be saved or if the submission fails.

The platform supports multiple feedback entries if a student attends multiple sessions. This use case ensures that the feedback process is efficient, transparent, and contributes to improving the counseling services in SAMS.[4]

**Table 3.13: Provide Feedback**

|                  |                                                                                                                                                                                                                                                                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 13                                                                                                                                                                                                                                                                                                                                                                |
| Use case name    | Provide Feedback                                                                                                                                                                                                                                                                                                                                                  |
| Actors           | Student                                                                                                                                                                                                                                                                                                                                                           |
| Description      | The student provides feedback after a counseling session.                                                                                                                                                                                                                                                                                                         |
| Trigger          | The student submits feedback for a completed counseling session.                                                                                                                                                                                                                                                                                                  |
| Pre-condition    | The student must be logged in.                                                                                                                                                                                                                                                                                                                                    |
| Post condition   | A counseling session must have been completed.                                                                                                                                                                                                                                                                                                                    |
| Normal flow      | <p>The feedback is successfully stored in the system. The counselor can view the feedback.</p> <p>Student logs into the system.</p> <p>Student navigates to the "Provide Feedback" section. Student selects the completed counseling session.</p> <p>Student enters feedback and submits it.</p> <p>The system saves the feedback and notifies the counselor.</p> |
| Alternative flow | The student can edit or delete feedback before final submission.                                                                                                                                                                                                                                                                                                  |
| Exceptions       | If the feedback submission fails, the system displays an error message.                                                                                                                                                                                                                                                                                           |
| Business rules   | <p>Students can only provide feedback for completed sessions.</p> <p>Feedback must be stored securely and visible to authorized personnel only.</p>                                                                                                                                                                                                               |

The use case below allows counselors to conduct and complete scheduled sessions. The system records session details, enables feedback submission, supports rescheduling, and

ensures only registered counselors can conduct sessions while maintaining records for future reference.

The Conduct Counseling Session use case allows registered counselors to carry out scheduled counseling sessions with students. The system verifies counselor authentication to ensure that only authorized personnel can conduct sessions. During the session, all details such as date, time, student name, and counseling type are recorded in the system. Counselors can take notes and update session outcomes directly in the platform. The system supports rescheduling of sessions if required, notifying both the student and the counselor about changes. After completing a session, students can submit feedback through the Provide Feedback module, which is automatically linked to the session record. The system logs all session activities and updates the student's counseling history for future reference.

**Table 3.14: Conduct Counseling Session**

|                  |                                                                                                                                                                                                  |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 14                                                                                                                                                                                               |
| Use case name    | Conduct Counseling Session                                                                                                                                                                       |
| Actors           | Counselor                                                                                                                                                                                        |
| Description      | The counselor conducts a counseling session with a student.                                                                                                                                      |
| Trigger          | The counselor initiates a session with a student.                                                                                                                                                |
| Pre-condition    | The counselor must be logged in.<br>The student must have requested counseling.                                                                                                                  |
| Post condition   | The session details are recorded in the system.<br>The student can provide feedback after the session.                                                                                           |
| Normal flow      | Counselor logs into the system.<br>Counselor navigates to the "Counseling Sessions" section. Counselor selects the scheduled session.<br>The session is conducted (either in-person or virtual). |
| Alternative flow | If the session needs to be rescheduled, the system updates the schedule.                                                                                                                         |
| Exceptions       | If the counselor is unavailable, the session is postponed.<br>If the system fails to save session details, an error message appears.                                                             |
| Business rules   | Only registered counselors can conduct sessions. Session records must be maintained for future reference.                                                                                        |

Receive Appointment Notification use case ensures students are notified about scheduled counseling sessions. The system sends notifications at least 24 hours before the appointment, logs them, allows manual checks if notifications are disabled, and handles errors for failed delivery or incorrect contact details.

The Receive Appointment Notification use case ensures that students are informed in advance about their scheduled counseling sessions. The system automatically generates notifications at least 24 hours before the appointment, providing sufficient time for students to prepare. Notifications include key details such as counselor name, session type, date, time, and location or mode of counseling. The system logs every notification sent, maintaining a history for reference and auditing. If a student has disabled notifications, the system allows manual checks so that they can still verify upcoming appointments. Error handling is built in to manage scenarios such as failed delivery, missing contact information, or system errors.

**Table 3.15: Receive Appointment Notification**

|                  |                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case id      | 15                                                                                                                                                                               |
| Use case name    | Receive Appointment Notification                                                                                                                                                 |
| Actors           | Student                                                                                                                                                                          |
| Description      | The student receives a notification for a scheduled counseling appointment.                                                                                                      |
| Trigger          | A counseling session is scheduled.                                                                                                                                               |
| Pre-condition    | The student must have requested counseling.<br>The counselor must have scheduled the session.                                                                                    |
| Post condition   | The student receives a notification.<br>The system logs the notification.                                                                                                        |
| Normal flow      | The counselor schedules a session.<br>The system records the appointment.<br>The system sends a notification to the student.<br>The student receives and views the notification. |
| Alternative flow | If the student has disabled notifications, they must check manually.                                                                                                             |
| Exceptions       | If the system fails to send the notification, an error log is generated.<br>If the student's contact details are incorrect, an error is displayed.                               |

|                |                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------|
| Business rules | Notifications must be sent at least 24 hours before the appointment.<br>Students must be able to view past notifications. |
|----------------|---------------------------------------------------------------------------------------------------------------------------|

### 3.3 Functional Requirements

The SAMS platform is designed with a structured set of modules to cater to different types of users within the university. The Student Module allows students to interact with the system by registering for events, submitting grievances, requesting counseling, and providing feedback, ensuring an engaging and supportive campus experience. The Admin Module enables administrators to manage student affairs efficiently, including event creation, grievance tracking, counselor assignments, and overall system oversight, ensuring smooth operations and accountability. The Mentor Module allows mentors or counselors to conduct sessions, review student requests, track counseling histories, and receive feedback, providing structured support and guidance to students. Together, these modules form a unified ecosystem that streamlines communication, improves transparency, and enhances student engagement across campus activities.

#### 3.3.1 Student Module

The Student Module allows students to register and log in using their email and password, with additional options for email verification and password recovery. Students can submit grievances through an online form, with each grievance assigned a unique case ID for easy tracking. They can monitor the status of their grievances, whether pending, in-progress, or resolved. Real-time notifications keep students updated about grievance progress, event participation, and counseling appointments. Additionally, students can register for university events via an interactive event calendar, request counseling sessions for academic, career, or personal guidance, and provide feedback after completing counseling sessions.

#### 3.3.2 Admin Module

The Admin Module enables administrators to manage user accounts by approving, deleting, or updating them. Admins can view, filter, and handle student grievances, approve or reject them, and assign review committees for resolution. They can create, update, and manage university events, add counselor details to allow students to select appropriate counselors, and send notifications regarding grievances, events, or counseling appointments. The system also supports generating reports and analytics for grievance statistics, student participation, and

overall system performance, while maintaining secure role-based authentication for students, admins, and counselors.

### **3.3.3 Mentor Module**

The Mentor Module allows mentors to log in securely, view academic-related grievances assigned to them, and provide responses or recommendations. Mentors can schedule and conduct academic counseling sessions, receive notifications when students request counseling or submit academic grievances, and communicate with students through the system for effective academic guidance.

## **3.4 Nonfunctional Requirements**

Nonfunctional requirements are those that establish parameters that may be used to evaluate the performance of a system. Nonfunctional requirements are the limitations to which the system will work.

### **3.4.1 Performance**

The app should be responsive and provide quick loading times, even during peak usage periods. Response times for actions such as event registration, message delivery, and content loading should be minimal. The system should be able to handle a large number of concurrent users without significant degradation in performance.

### **3.4.2 Security**

Data privacy and protection should be ensured through robust security measures, including encryption of sensitive information and secure authentication mechanisms. User authentication and authorization processes should be implemented securely to prevent unauthorized access to user accounts and data. The app should comply with relevant data protection regulations and standards to safeguard user privacy.

### **3.4.3 Availability**

This system must be always accessible, 24 hours a day, seven days a week.

### **3.4.4 Ease of Use**

The user interface should be intuitive, user-friendly, and accessible to students of varying technical abilities. Navigation within the app should be straightforward, with clear labeling and intuitive design patterns. The app should support multiple languages and accommodate diversity.

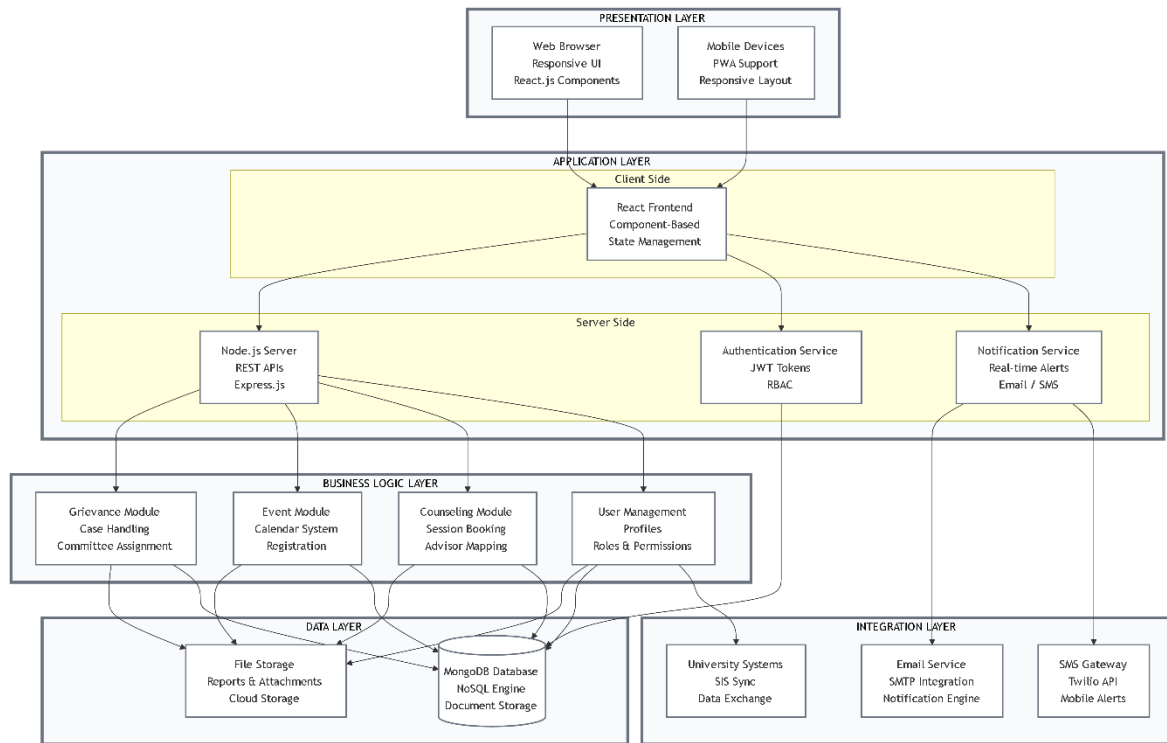
# **Chapter 4**

## **Design and Architecture**

## 4 Design and Architecture

### 4.1 System Architecture

The structure of the system explains the working of the system and describes the components of the SAMS, their relationships, and how they interact with each other, with this architecture.



**Figure 4.1: SAMS Architecture**

The SAMS system starts with users accessing it through web browsers or mobile devices, where the interface is built using React.js to ensure smooth and responsive design across all screen sizes. The frontend handles user actions and manages data using components and state management. On the backend, a Node.js server with Express.js handles requests and connects the frontend to the database through REST APIs. User login and access control are managed using JWT tokens and role-based permissions. A notification service sends real-time alerts via email or mobile. The core of the system includes modules for handling grievances, reviews, counseling, reporting, and user management—each allowing users to create, view, and update records. All

data is stored in MongoDB, a fast NoSQL database, while files like reports and images are saved separately.

The system also connects with university tools for data sharing, and alerts are sent through email servers, SMS gateways, and mobile notifications to keep users informed.

## **4.2 Data Representation**

In SAMS, data is stored using a structured and well-organized database design that supports efficient management of grievances, events, counseling requests, and user information. Each module such as Grievance, Event, Counseling Session, and User uses predefined schemas to ensure data consistency, accuracy, and easy retrieval across the system. This structured representation allows the platform to process submissions, update statuses, and manage notifications smoothly, ensuring that information is encoded, stored, and retrieved efficiently for both students and administrators.

The Event and Counseling Session modules in SAMS use structured database schemas to efficiently manage information. The Event module records details such as event ID, name, date, time, venue, organizer, and registered participants, enabling easy filtering, sorting, and management of events for both students and administrators. Similarly, the Counseling Session module tracks session ID, student and counselor IDs, date and time, session type, and status, allowing students to schedule appointments, receive automated reminders, and enabling counselors to access session histories for informed guidance.

User data is handled through a dedicated User module, which stores essential information such as user ID, name, email, password role, and contact information. By using predefined schemas, the system ensures that all user-related information remains consistent and easily accessible for authentication, authorization, and personalized interactions across the platform.

The structured nature of the SAMS database enables efficient data processing. When a student submits a grievance or registers for an event, the system automatically updates the respective tables and triggers notifications to relevant parties. Status updates, whether for grievances, event participation, or counseling appointments, are stored and reflected in real-time, ensuring that both students and administrators have accurate, up-to-date information at all times.



Furthermore, this organized representation of data allows for easy retrieval and reporting. Administrators can generate reports to track the number of grievances submitted, pending, or resolved over a certain period, evaluate event attendance trends, or monitor counseling session schedules. By adhering to standardized schemas across all modules, SAMS minimizes the risk of data duplication, inconsistency, and errors, while also optimizing query performance.

Overall, the database design in SAMS not only ensures the safe storage of information but also supports smooth operations, timely notifications, and reliable access to data.

### **4.3 Process Flow**

The Process Flow Representation illustrates the step-by-step sequence of actions and interactions between different components in the SAMS platform. It provides a visual overview of how students, admins, counselors, and the system work together to manage grievances, events, and counseling services.

#### **Grievance Management:**

In SAMS, when a student submits a grievance through the platform, the system records it and assigns a unique ID for tracking.

The system then notifies the administrator about the new submission and updates the grievance status as actions are taken, while sending regular status updates to the student. Administrators can view and filter submitted grievances, assign a committee to review them, and ultimately approve or reject each grievance based on a thorough evaluation.

#### **Event Management:**

In SAMS, administrators create and manage events by entering all relevant details and setting dates. Students can view the event calendar, browse available events, and register for those of interest, receiving notifications about upcoming events.

The system securely stores all event information, confirms student registrations, and sends automated reminders 24 hours before each event to ensure smooth participation.

## **Student Counseling:**

In SAMS, students can request counseling sessions, receive appointment notifications, attend the sessions, and provide feedback afterward. Administrators manage the process by viewing counseling requests and adding counselor details to the system.

Counselors conduct the sessions and mark them as completed. The system records all session details, updates records, notifies students about session updates, and logs feedback for future reference, ensuring an organized and efficient counseling management process.

This process flow ensures that each role—student, admin, counselor, and system—has clearly defined tasks and interactions. The SAMS platform is designed to streamline student support services, making grievance handling, event participation, and counseling smooth and efficient for all users.

The integration of automated notifications and record-keeping reduces administrative workload while keeping students informed at every step. Feedback collected from counseling sessions is stored systematically, enabling counselors and administrators to monitor trends, evaluate service effectiveness, and implement improvements over time. This structured approach ensures that the quality of student support remains consistent and reliable.

Moreover, the SAMS platform enhances accessibility by allowing students to request and manage their counseling sessions remotely, without needing to visit administrative offices physically. This convenience encourages more students to engage with counseling services, promoting mental well-being and academic support. The digital workflow also facilitates transparency, as students can track the status of their requests in real-time, and administrators can ensure that no session requests are overlooked.

Finally, the counseling module, as part of the larger SAMS ecosystem, demonstrates how technology can simplify and improve student affairs management. It not only improves operational efficiency.

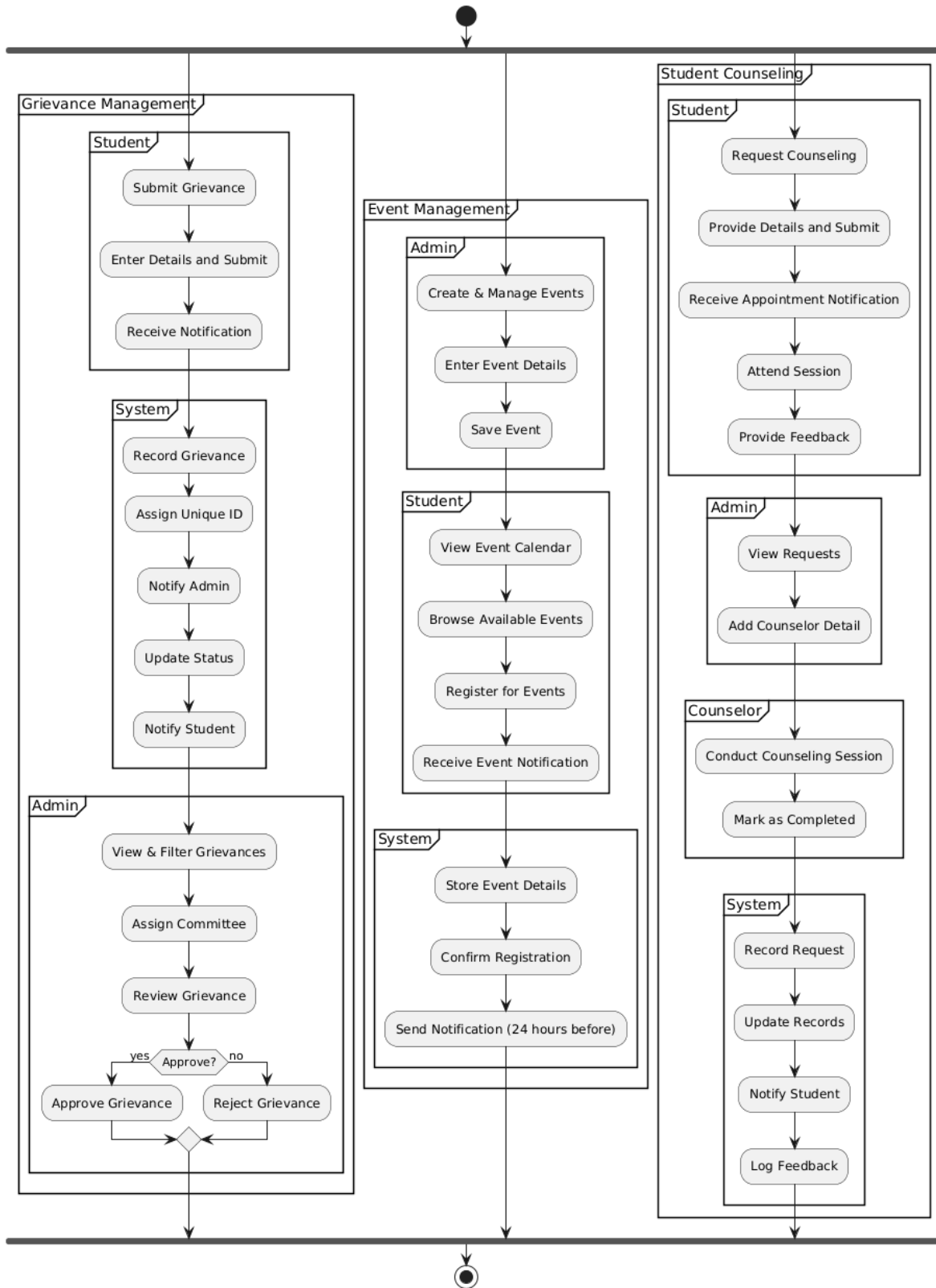


Figure 4.2: SAMS Process Flow

## 4.4 Sequence Diagram

These figure shows how the system will work through sequence diagrams, task after task sequence wise. It gives a complete description of the working of SAMS.

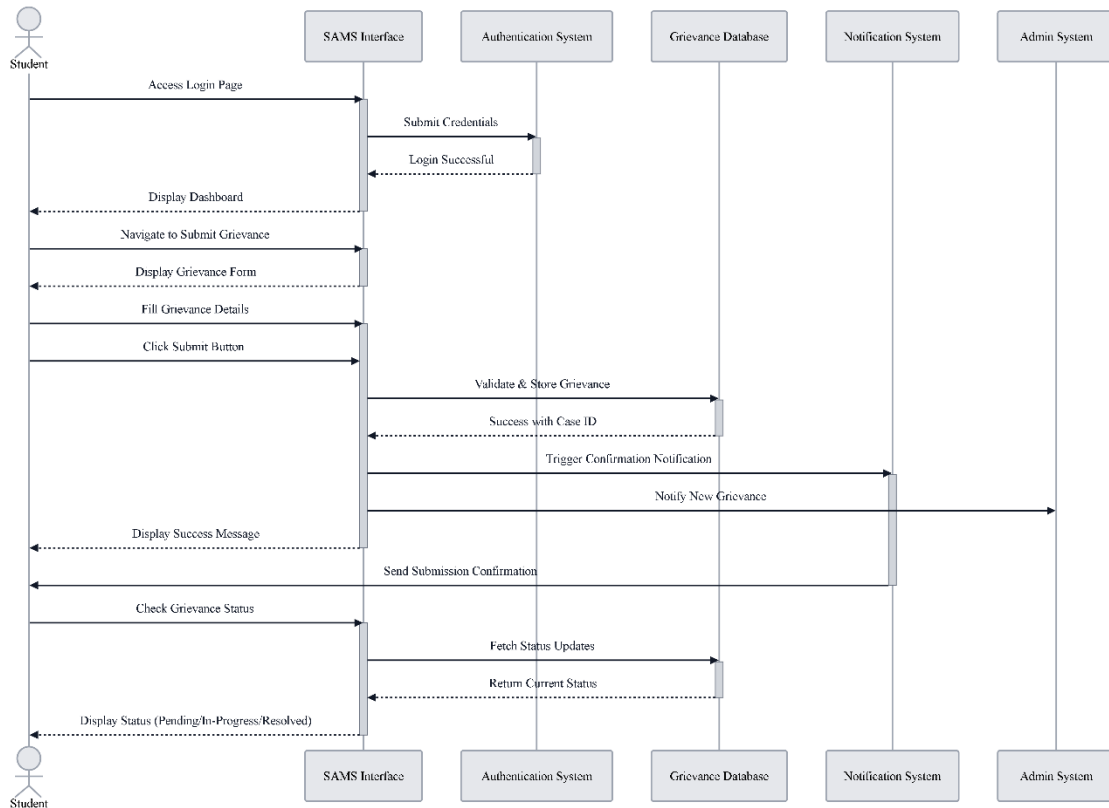
### **Phase1: Student Grievance Management**

The diagram illustrates how a student submits a grievance using the SAMS platform. The process begins when the student accesses the login page through the SAAS Interface and enters their credentials. These credentials are sent to the Authentication System for verification. Once the login is successful, the student is directed to the dashboard and navigates to the grievance submission section. The student fills out the grievance form and clicks submit. The SAAS Interface validates the form and sends the grievance to the Grievance Database, which stores the request and notifies the Notification System.

A confirmation message is then displayed to the student. Later, the student can check the status of their grievance, and the SAAS Interface retrieves the current status from the Grievance Database. The system shows whether the grievance is pending, in progress, or resolved. This flow ensures that the student is guided step-by-step and kept informed throughout the grievance process.

The diagram illustrates how a student submits a grievance using the SAMS platform. The process begins when the student accesses the login page through the SAMS interface and enters their credentials. These credentials are sent to the Authentication System for verification. Once the login is successful, the student is directed to the dashboard, which provides access to various modules, and navigates to the grievance submission section.

The student then fills out the grievance form, providing details such as the nature of the issue, relevant dates, and any supporting documents. Upon clicking submit, the SAMS interface validates the form to ensure all mandatory fields are completed and that the data is in the correct format. Once validated, the grievance is sent to the Grievance Database, where it is securely stored. Simultaneously, the Notification System is triggered to inform the student that their grievance has been received and assigned a unique case ID for tracking.

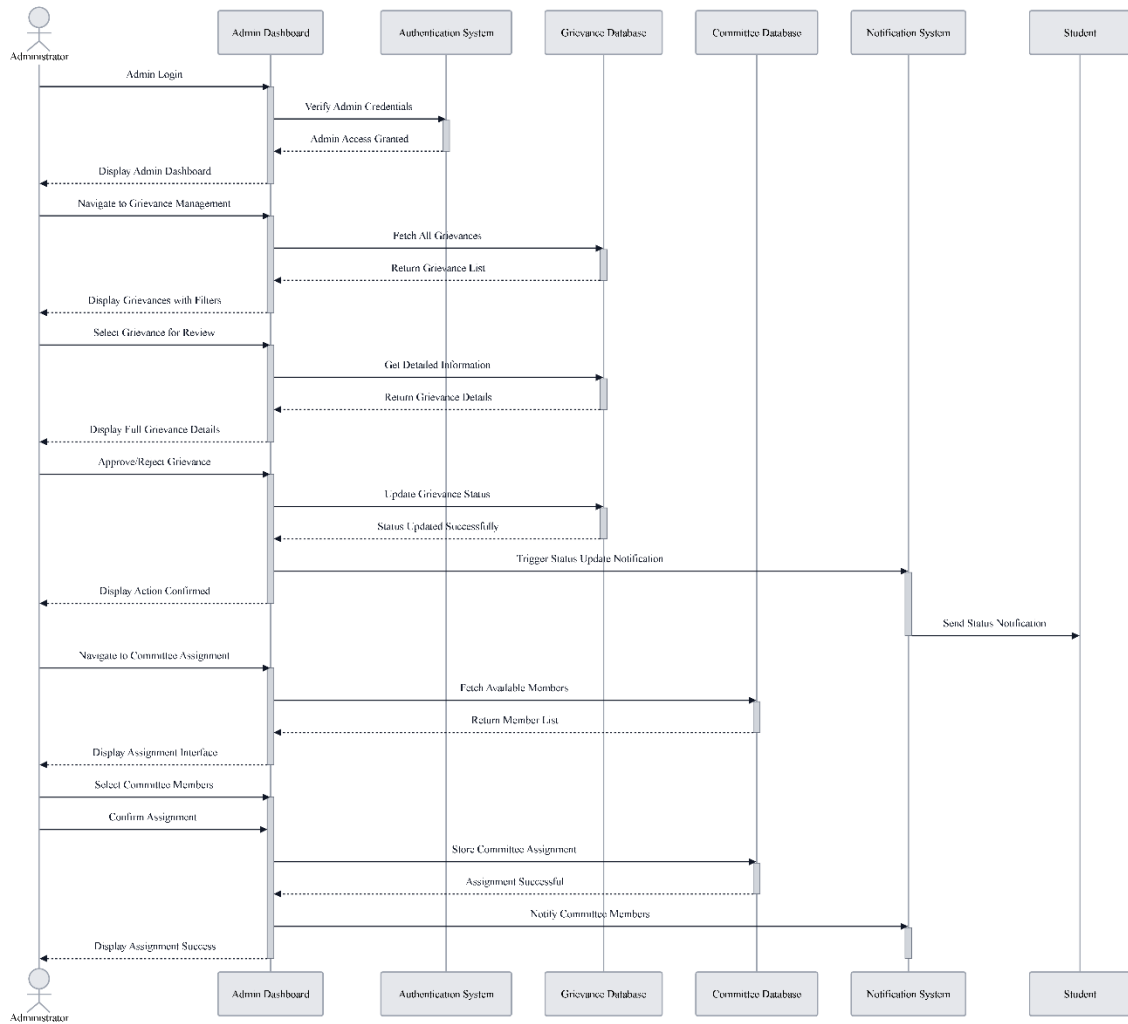


**Figure 4.3: Student Grievance Management**

## Phase2: Admin Grievance Management

The diagram outlines the full process of how an Administrator manages student grievances and assigns them to committees using the SAMS platform. It begins with the Administrator logging into the Admin Dashboard, which verifies credentials through the Authorization System. Once access is granted, the Administrator navigates to the grievance dashboard and views submitted grievances.

Once a grievance is assigned, the system records the action in the Grievance Database and sends notifications to both the assigned committee and the student. Committees update the status of the grievance as they review and resolve it, while the Administrator monitors progress through the dashboard. Each update is reflected in the system, ensuring students are informed of whether their grievance is pending, in progress, or resolved.



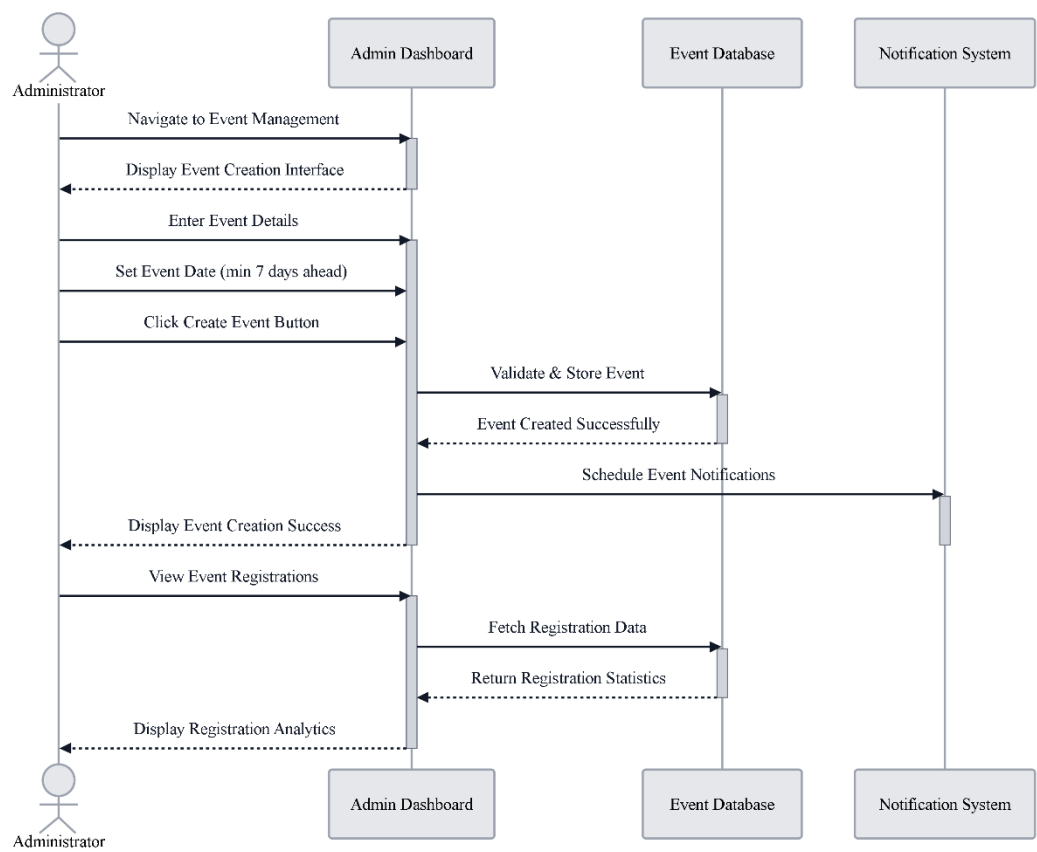
**Figure 4.4: Admin Grievance Management**

The system allows the Administrator to review grievance details, approve or reject them, and update their status in the Grievance Database. After confirming the status, the Administrator accesses the Committee Database to view available committee members and assign students accordingly. Once the assignment is successful, the Admin Dashboard triggers a notification through the Notification System, which sends a status update to the student. Finally, the system confirms the assignment and logs the Administrator out. This sequence ensures that grievances

are handled efficiently, committee assignments are properly managed, and students are kept informed throughout the process.

**Phase3: Admin Event Management:**

The sequence diagram outlines how an Administrator interacts with the SAMS platform to create and manage events. The process begins when the Administrator accesses the Event Management section through the Admin Dashboard, which then displays the Event Creation Interface. The Administrator enters event details and sets a date that must be at least seven days ahead. Upon clicking the "Create Event" button, the Admin Dashboard sends the event data to the Event Database for validation and storage. Once the event is successfully created, the Event Database schedules notifications through the Notification System to alert users. The Administrator can then view event registrations via the dashboard, which fetches registration data from the Event Database. This data includes statistics and analytics, which are displayed to the Administrator.



**Figure 4.5: Admin Event Management**

The Administrator can then view event registrations via the dashboard, which fetches registration data from the Event Database. This data includes statistics and analytics, which are displayed to

help the Administrator monitor participation and engagement. This sequence ensures a smooth and organized workflow for planning, notifying, and tracking events within the SAMS platform.

#### **Phase4: Student Event Management**

The sequence diagram illustrates how a student interacts with the SAMS platform to register for an event. The process begins when the student navigates to the Event Calendar through the SAMS Interface.

The interface then fetches upcoming events from the Event Database and displays them to the student. Upon selecting a specific event, the system requests detailed information from the Event Database, which is then shown to the student. When the student clicks the register button, the SAMS Interface checks for available slots in the Registration Database.

If slots are available, the system creates a registration record and confirms the registration. Upon selecting a specific event, the system fetches more detailed information from the Event Database, ensuring that the student has all the necessary details before registering. When the student clicks the Register button, the SAMS interface verifies slot availability in the Registration Database. If seats are available, a registration record is created, and the student receives a confirmation message.

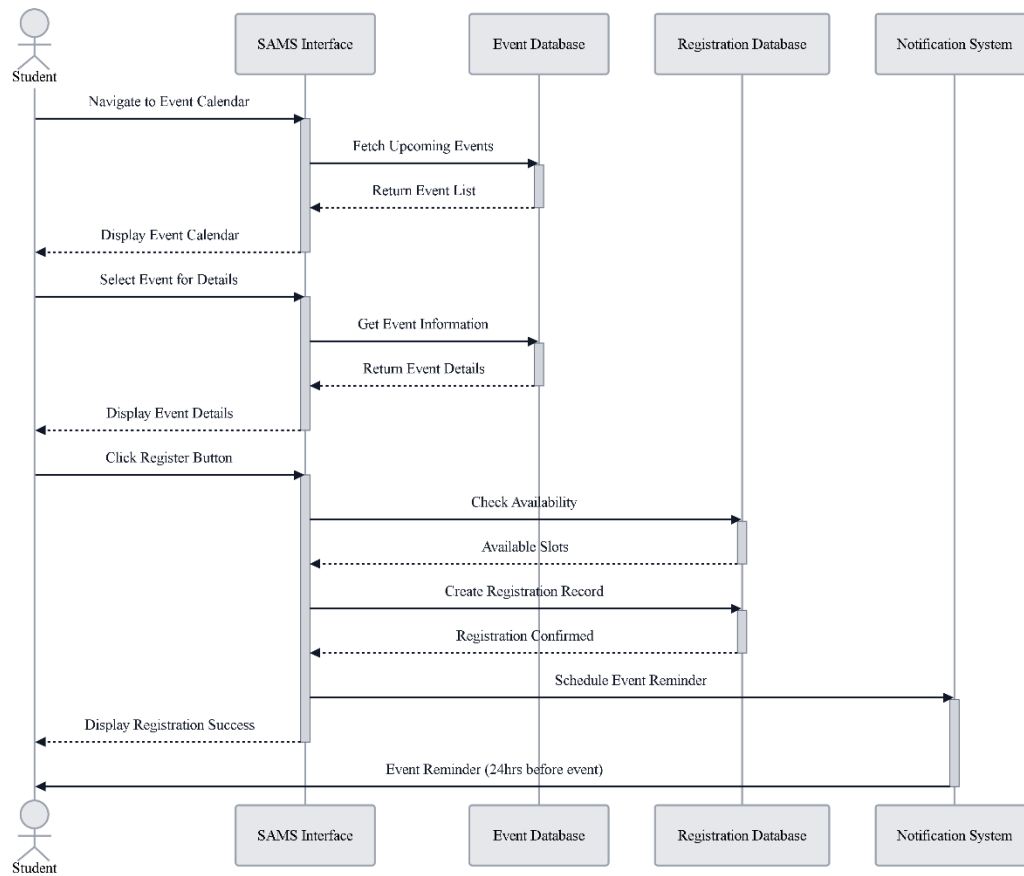
Additionally, the system schedules a reminder through the Notification System, which ensures that the student is notified prior to the event. This structured process ensures smooth, transparent, and efficient event registration, while also keeping students informed and engaged throughout the workflow.

This structured workflow not only ensures smooth and efficient event registration but also enhances student engagement by providing timely updates and confirmations. By centralizing event information, automating slot verification, and sending notifications, SAMS reduces manual errors and administrative overhead while providing students with a seamless, user-friendly experience.

Overall, this module strengthens the connectivity between students and campus activities, encouraging active participation and better utilization of university resources.

Following this, the SAMS Interface schedules a reminder with the Notification System, which confirms the setup.





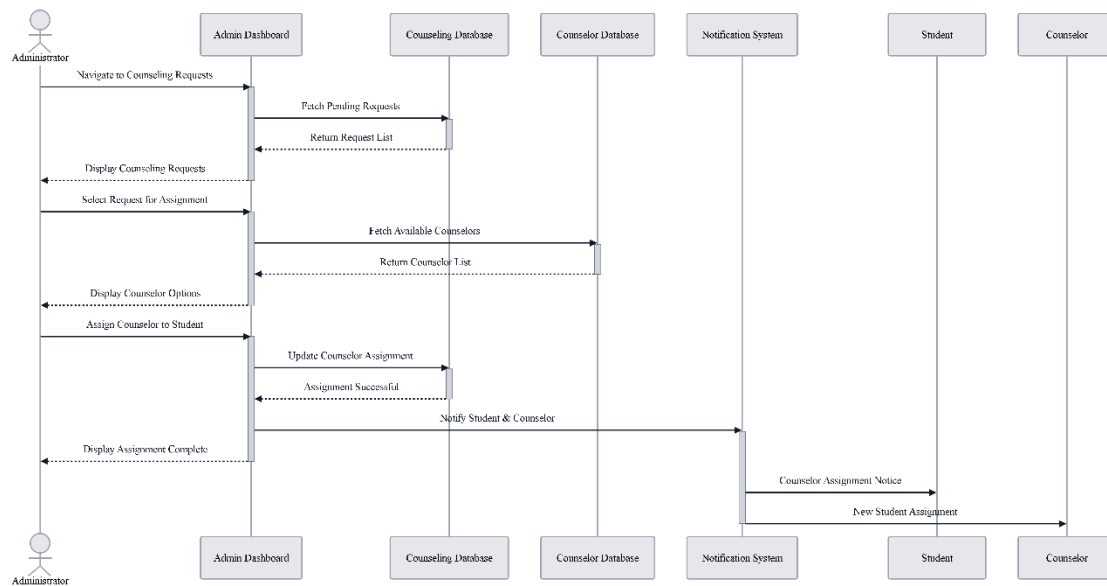
**Figure 4.6: Student Event Management**

Finally, the student receives a success message indicating that the registration is complete. This sequence ensures a smooth and responsive experience for students, allowing them to explore, register, and stay informed about events efficiently.

### Phase5: Admin Counseling Management

The diagram outlines the complete workflow for assigning a counselor to a student within the SAMS platform. It begins with the Administrator logging into the Admin Dashboard and navigating to the Counseling Requests section. The dashboard displays all pending requests, and the Administrator selects one for assignment. Available counselor options are fetched from the Counselor Database and displayed. Once a counselor is chosen, the Admin Dashboard updates the assignment in the Counseling Database. Simultaneously, the Notification System sends alerts

to both the student and the assigned counselor, informing them of the new appointment. The Counselor Database also updates its records to reflect the new assignment. This process ensures that counseling requests are handled efficiently, with clear communication between the system, students, and counselors, streamlining the support process for student well-being.



**Figure 4.7: Admin Counseling Management**

## Phase6: Student Counseling Management

The diagram outlines the complete journey a student follows to request and receive a counseling session through the SAMS platform. It begins when the student accesses the Counseling Session section via the SAAS Interface, which displays available counseling information. The student selects the type of counseling and submits their request details. This request is sent to the Counseling Database, which generates a unique Request ID and stores the information. The Counseling Database then alerts the Notification System, which sends a confirmation notification back to the student. The student can later check the appointment status, prompting the SAAS Interface to query the Counseling Database. Once the request is processed, the Counseling Database triggers another notification to inform the student about the counselor assignment. Simultaneously, the Admin System receives the assignment details and assigns the student to a counselor.

The Counseling Database then sends the counselor assignment and schedule information to the SAAS Interface, which displays the final appointment details to the student.

This sequence ensures a smooth and transparent counseling process, keeping both students and counselors informed at every step.

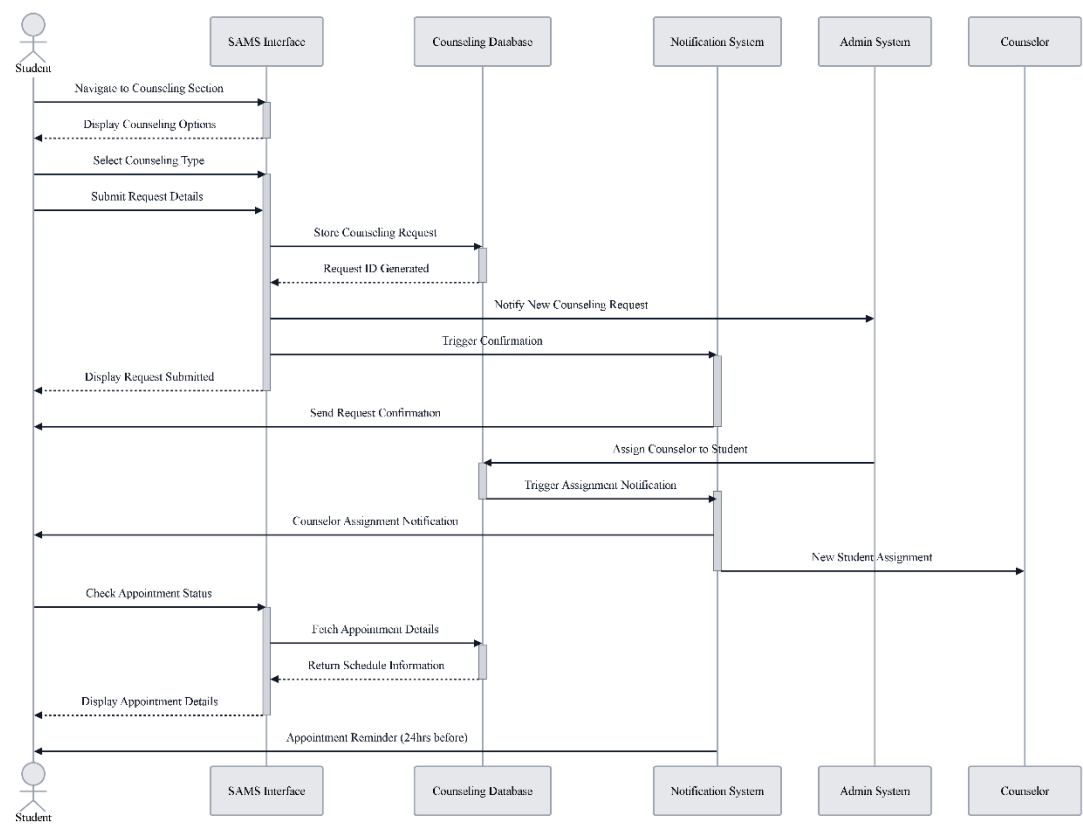


Figure 4.8: Student Counseling Management

Phase7: Notification System

This diagram shows how the SAMS system sends notifications to students through email and SMS. It starts when the System triggers a notification event, which is passed to the Notification Engine. The engine adds the notification to a Message Queue, which confirms that it has received the message. The Notification Engine then processes the queue, fetches the notification data, and sends it to the Email Service.

The Email Service delivers the email to the student and reports back with the delivery status. Next, the Notification Engine sends the same message to the SMS Gateway, which delivers the SMS to the student and also returns the delivery status. Finally, the Notification Engine updates the delivery status in the SAVS System to confirm that the notifications were successfully sent. This process ensures that students receive timely alerts through multiple channels, keeping them informed and engaged.

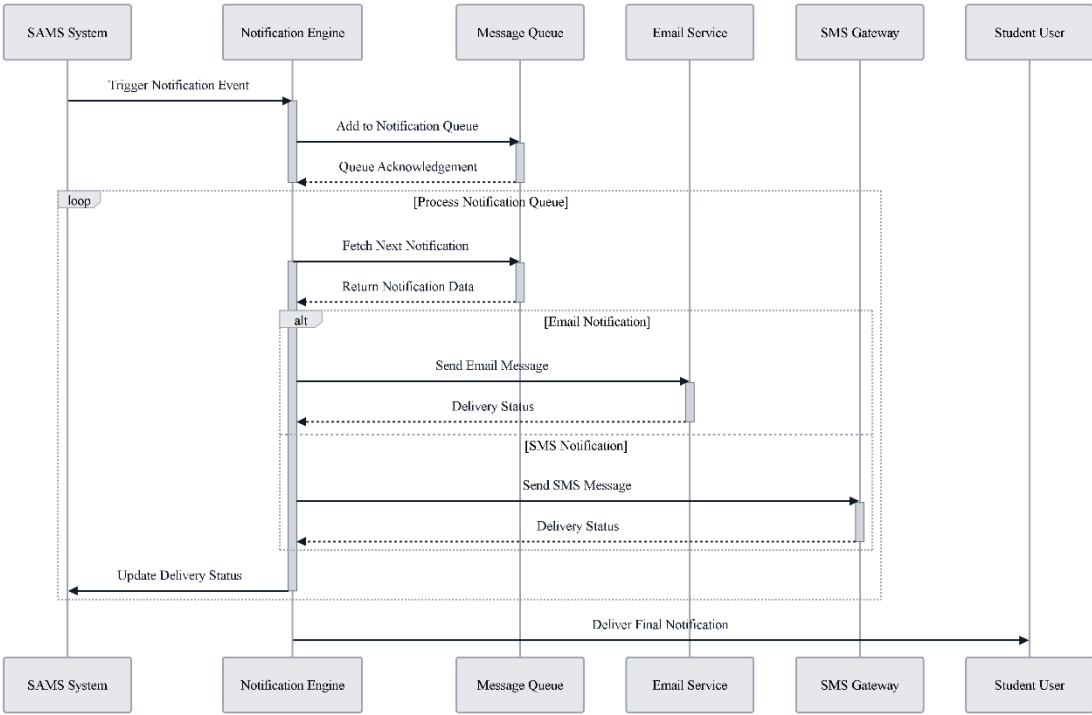
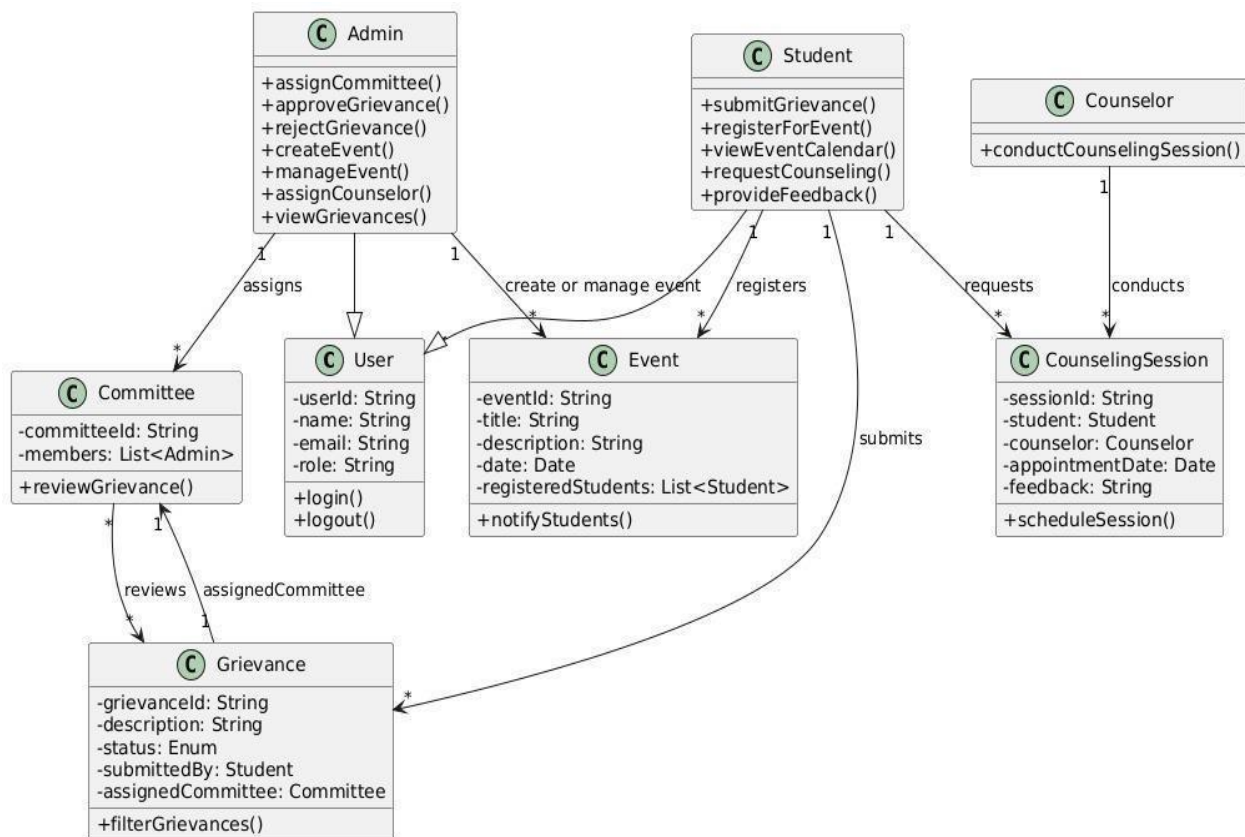


Figure 4.9:Notification System

4.5 Class Diagram

The class diagram represents the structure of the SAMS showing how different parts of the system are connected and what each part does. At the center is the User class, which holds basic information like user ID, name, email, and role. This class acts as a parent to more specific roles such as Admin, Student, and Counselor, each with their own responsibilities. The Admin class is responsible for managing grievances and events. Admins can assign committees, accept or reject grievances, review them, and view all grievance records. They also create and manage events,

linking them to the Event class. The Student class allows students to register for events, view the event calendar, request counseling, and submit grievances. These actions connect students to the Event, Grievance, and Counseling Session classes.



**Figure 4.10: SAMS Class Diagram**

The Counselor class is focused on conducting counseling sessions. Each session is represented by the Counseling Session class, which includes details like session ID, date, counselor, student, feedback, and a method to schedule the session. The Committee class is used to review grievances. It contains a committee ID and a list of Admin members. Grievances are assigned to committees, and the committee reviews them. The Event class stores event details such as ID, title, description, date, and a list of registered students. It also includes a method to notify students about events. The Grievance class holds grievance information like ID, description, status, the student who submitted it, and the assigned committee. It also includes a method to filter grievances. Finally, the Counseling Session class connects students and counselors, storing

session details and feedback. All these classes are linked through relationships showing how users interact with the system and how data flows between components.

#### **4.5.1 Summary:**

This diagram illustrates the structure and relationships within a social platform where users interact with various components. Admins manage the platform's activities, while users engage with the content and each other. The system is designed to facilitate communication.

# **Chapter 5**

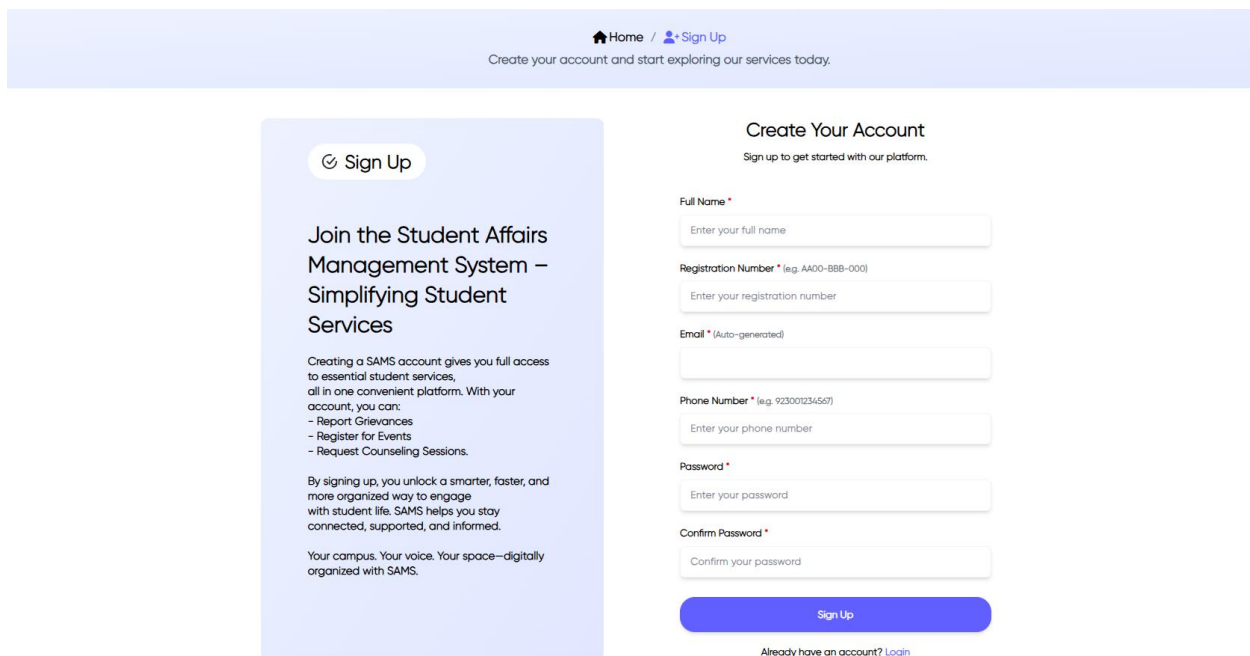
## **Implementation**

## 5 Implementation

The following section presents the complete implementation of the SAMS system, demonstrating how each module functions within the platform.

### 5.1 User interface

The user interface SAMS is designed to be simple, intuitive, and visually appealing. Below are the main screens:



The screenshot displays the 'Sign Up' page of the SAMS system. At the top, a navigation bar includes links for 'Home' and 'Sign Up', along with the text 'Create your account and start exploring our services today.' The main content area is split into two columns. The left column, titled 'Join the Student Affairs Management System – Simplifying Student Services', describes the benefits of creating a SAMS account, such as access to essential student services and the ability to report grievances, register for events, and request counseling sessions. The right column, titled 'Create Your Account', contains a form with fields for 'Full Name', 'Registration Number', 'Email', 'Phone Number', 'Password', and 'Confirm Password'. A 'Sign Up' button is located at the bottom of the form, and a link for 'Already have an account? Login' is provided below it.

**Figure 5.1: Sign up Page**

The Sign-Up Page allows new students to create their accounts in the SAMS system. The left section introduces the platform and highlights services such as advisor support, academic record access, grievance submission, event registrations, and counseling appointments.

On the right side, a structured sign-up form is provided where students enter their name, registration number, email, phone number, and password, along with a confirm-password field for verification. A Sign Up button completes the registration process, and a quick link is available for users who already have an account, ensuring an easy and organized onboarding experience.



This page is designed to make the first interaction simple and welcoming for students. It ensures that all essential details are collected in one go without confusion. By combining clear instructions with a user-friendly layout.

Home / Login

Access your account to view your profile and continue your journey with us.

Login

Welcome to Student Affairs Management System

We empower students by offering a user-friendly platform to report grievances, register for events, and request counseling sessions.

Welcome back!

Please login to access your account.

Registration Number \* (eg. AA00-BBB-000)

Enter your registration number

Password \*

Enter password

Login

Admin Access

Don't have an account yet? [Sign up](#)

Forgot Password? [Reset Now](#)

**Figure 5.2: Login Page**

The Login Page provides students access to their SAMS accounts. The left side welcomes users with a brief introduction and highlights the system's seamless student services. On the right, the login form allows students to enter their registration number and password, supported by a clean and user-friendly interface. Options for admin login, account creation, and password recovery are also available. After successful login, the student is directed to the dashboard, ensuring secure access and smooth navigation.

The Login Page serves as the main entry point for students to access their SAMS accounts, ensuring secure and authenticated access to all system features. The left section of the page offers a warm welcome message along with a short introduction that highlights the platform's goal of providing smooth, centralized, and efficient student services. On the right side, a clean and structured login form is presented where students can enter their registration number and password. The interface is designed to be simple, accessible, and visually organized to avoid confusion.



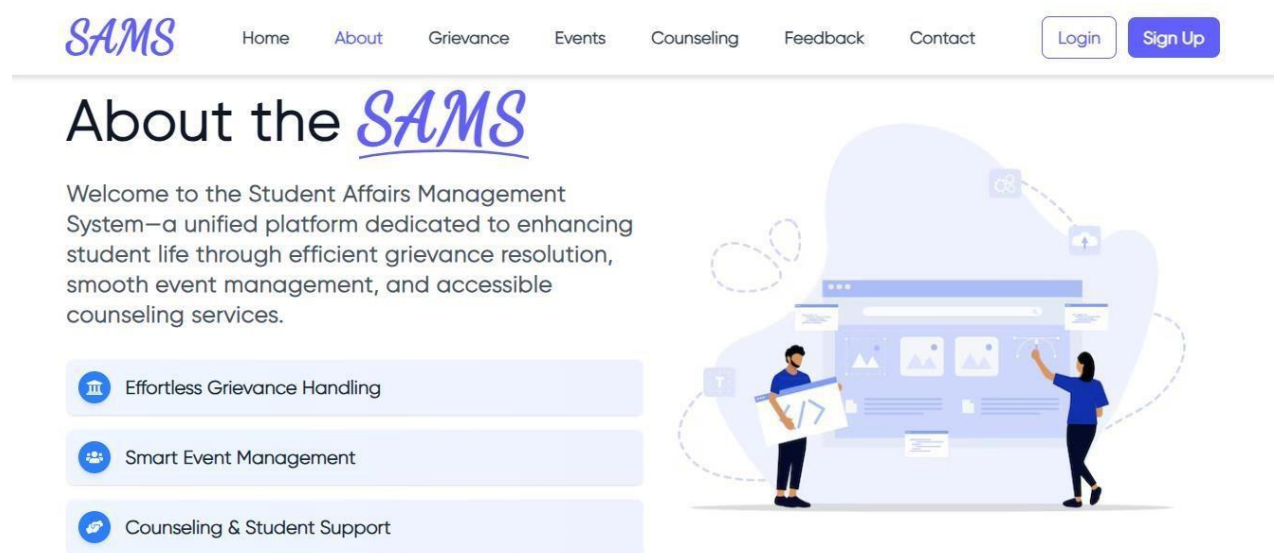
**Figure 5.3: Home Page**

The Homepage serves as the landing page of SAMS, presenting a visual introduction and essential navigation options. A top navigation bar provides links to major sections like Grievance, Events, and Counseling, while the background includes an image of the university building to enhance visual appeal. The center displays a welcome message along with the system name, emphasizing academic excellence and student support.

Additional elements like a Learn More button and slideshow navigation dots enrich the user experience. Login and Sign-Up buttons appear at the top-right corner, making this page both informative and inviting.

The Homepage serves as the central landing page of SAMS, offering a visually appealing introduction and easy navigation for all users. At the top, a clean navigation bar provides quick access to major sections such as Grievance, Events and Counseling, ensuring smooth movement throughout the platform. The background features a professional image of the university building, adding authenticity and enhancing visual engagement. In the center, a welcoming message along with the system name highlights SAMS's dedication to academic excellence and student support.

Additional interactive elements, such as a Learn More button and slideshow navigation dots, help users explore key features. Login and Sign-Up buttons are clearly placed at the top-right corner, making the homepage both informative and user-friendly, guiding students effortlessly into the system.



**Figure 5.4: About Page**

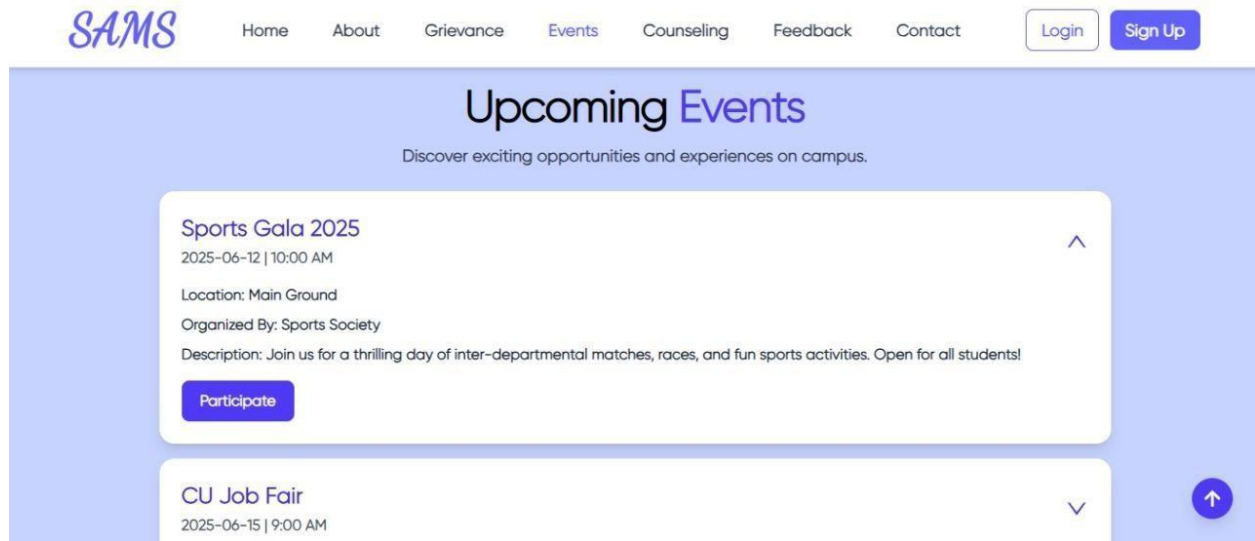
The About Page explains the core purpose and features of the SAMS platform. With a clear heading titled About the SAMS, the page outlines how the system enhances student life through efficient grievance handling, smart event management, and counseling support services.

Three icons visually represent these features, making the content easy to understand. The page includes navigation links, Sign-Up and Login buttons, and an illustration symbolizing digital student engagement. Overall, it provides a clear and visually appealing introduction to the system's benefits.

This page not only highlights the functional aspects of SAMS but also builds trust by showing students that their needs are prioritized. The combination of text and visuals makes the information accessible to all users, regardless of their technical background.

By presenting the system as supportive and student-focused, the About Page encourages new users to explore further and motivates them to register and actively engage with the platform.

In addition to showcasing the system's features, the page builds credibility by emphasizing how SAMS prioritizes student needs through transparency, convenience, and user-friendly design. The blend of visuals and concise explanations makes the information easy to follow for students of all technical levels.



**Figure 5.5: Event Page**

The Upcoming Events Page lists upcoming campus activities where students can participate. A bold heading and subheading create excitement, followed by event cards such as Sports Gala 2025 and CU Job Fair, each showing the date, time, venue, organizer, and a brief description. Each event includes a Participate button for quick registration. The navigation bar and sign-in options remain accessible at the top, while the clean interface promotes student engagement and easy navigation.

The Events Page provides students with an organized and visually engaging space to explore all upcoming university events. At the top, a clear heading introduces the purpose of the page, ensuring students immediately understand that this section is dedicated to event discovery and participation. A clean navigation bar remains accessible for easy switching between different sections of the SAMS platform.

The page displays a grid or list of events, each represented through event cards containing the title, date, location, and event description. These cards may also include images or icons to make browsing more appealing and intuitive. Students can easily scroll through academic workshops, seminars, competitions, cultural activities, and society-led programs.

A Register Now button is available on each event card, allowing students to sign up with just one click. Once registered, the system stores their participation and ensures they receive timely reminders and notifications—especially the 24-hour pre-event reminder to help them.

## Submit Your Grievance

Please fill out the form below to report your grievance. We are here to help you resolve your issues.

|                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Applicant Name</b></p> <div style="border: 1px solid #ccc; padding: 5px; display: flex; justify-content: space-between;"><span>Enter your full name</span></div>                                                                                                                         | <p><b>Registration Number</b></p> <div style="display: flex; justify-content: space-between;"><div style="border: 1px solid #ccc; padding: 5px; flex: 1;">SP25 ▼</div><div style="border: 1px solid #ccc; padding: 5px; flex: 1;">BBA ▼</div><div style="border: 1px solid #ccc; padding: 5px; flex: 1;">Roll No </div></div> |
| <p><b>Phone Number</b></p> <div style="border: 1px solid #ccc; padding: 5px; display: flex; justify-content: space-between;"><span>Enter your phone number</span></div>                                                                                                                        | <p><b>University Email</b></p> <div style="border: 1px solid #ccc; padding: 5px; display: flex; justify-content: space-between;"><span>Enter your university email</span></div>                                                                                                                                               |
| <p><b>Issue Type</b></p> <div style="border: 1px solid #ccc; padding: 5px;"><span>Select issue type ▼</span></div>                                                                                                                                                                                                                                                              | <p><b>Date of Issue</b></p> <div style="border: 1px solid #ccc; padding: 5px; display: flex; justify-content: space-between;"><span>mm/dd/yyyy</span></div>                                                                                                                                                                   |
| <p><b>Description of the Issue</b></p> <div style="border: 1px solid #ccc; padding: 10px; min-height: 60px;"><span>Describe your issue in detail...</span></div>                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Evidence (Optional)</b></p> <div style="border: 1px solid #ccc; padding: 5px; display: flex; justify-content: space-between;"><span>Choose Files No file chosen</span></div> <p style="font-size: 0.8em; margin-top: 5px;">You can upload images, videos, or documents (if any).</p> |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><a href="#" style="color: #4a7ebb; text-decoration: none;">+ Add Another Person</a></p>                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Any Other Information (Optional)</b></p> <div style="border: 1px solid #ccc; padding: 10px; min-height: 40px;"><span>Provide any additional information...</span></div>                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><input type="checkbox"/> I confirm that all the information provided is correct.</p>                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <div style="background-color: #4a7ebb; color: white; padding: 10px 20px; display: inline-block; border-radius: 5px;">Submit Grievance</div>                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Figure 5.6: Grievance Page**

The Grievance Submission Form allows students to formally report their issues. The page begins with a welcoming heading and reassuring subtitle. The form collects essential details such as name, registration number, program, roll number, phone number, and university email.

Students can select the issue type, enter the date, and describe the problem in detail. Options for uploading supporting evidence and adding another involved person are provided.

A confirmation checkbox ensures data accuracy, and the “Submit Grievance” button sends the report, making grievance handling structured and efficient.

The Grievance Submission Form provides students with a structured and reliable way to formally report their concerns or issues. The page begins with a clear, welcoming heading followed by a reassuring subtitle that encourages students to express their problems confidently.

## Book Your Counseling Session

Take the first step towards your future. Fill out the form below to schedule your personalized counseling session.

|                                                               |                                               |                                  |                                       |
|---------------------------------------------------------------|-----------------------------------------------|----------------------------------|---------------------------------------|
| Full Name                                                     | Registration Number                           |                                  |                                       |
| <input type="text" value="Enter your name"/>                  | <input type="text" value="SP25"/>             | <input type="text" value="BBA"/> | <input type="text" value="Roll No."/> |
| Counseling Type                                               | Counselor Name                                |                                  |                                       |
| <input type="text" value="Select counseling type"/>           | <input type="text" value="Select counselor"/> |                                  |                                       |
| Day                                                           | Time Slot                                     |                                  |                                       |
| <input type="text" value="Select available day"/>             | <input type="text" value="Select time slot"/> |                                  |                                       |
| Issue Description                                             |                                               |                                  |                                       |
| <input type="text" value="Please describe your concerns..."/> |                                               |                                  |                                       |
| <input type="checkbox"/> I agree to the terms and conditions  |                                               |                                  |                                       |
| <input type="button" value="Schedule Counseling Session"/>    |                                               |                                  |                                       |

Figure 5.7: Counseling Page

The form gathers all necessary details. Students can choose the grievance category, enter the date of the incident, and describe the problem in detail to ensure accurate understanding.[5]

The Counseling Booking Form enables students to schedule personalized counseling sessions. The heading clearly defines the purpose, and the form includes fields for name, registration number, counseling type, chosen counselor, preferred date and time, and a description of the issue.

A terms-and-conditions checkbox ensures responsible submissions, and the Schedule Counseling Session button finalizes the request. This well-structured form supports flexibility and easy booking, strengthening academic and emotional support services.

The Counseling Booking Form enables students to conveniently schedule personalized counseling sessions through the SAMS platform. The heading clearly communicates the purpose of the form, ensuring users understand its function at a glance. Overall, this well-structured and user-friendly form enhances accessibility, ensures smooth appointment scheduling

The screenshot displays the 'Contact Us Form' on the SAMS platform. The navigation bar at the top includes the SAMS logo and links to Home, About, Grievance, Events, Counseling, Feedback, and Contact. The main heading is 'Contact Us Form' with a subtext: 'Have questions or need help? Our team is ready to assist you. Reach out today and let's start the conversation.' The form is divided into two panels. The left panel contains contact information: Address (Comsats University Rd, off GT Road, Sahiwal, Punjab 57000), Phone (Tel: +92-040-4305001, +92-040-4305002; Fax: +92-040-4305006), and Email (sams@cuisahiwal.edu.pk, info@cuisahiwal.edu.pk). The right panel is titled 'Send Us Message' and includes fields for Email, Registration No., Contact Number, and a Message box, followed by a 'Submit the message' button.

**Figure 5.8: Contact Us Page**

The Contact Us page provides students an easy way to reach the SAMS team. The left panel displays the system's physical address, phone number, and official email. On the right, a clean

form allows students to enter their email, registration number, phone number, and message. The “Submit the message” button sends inquiries directly.[6]

**Figure 5.9: Feedback Page**

The Feedback Form collects student opinions to improve SAMS services. The left section gives a warm welcome and brief introduction to encourage participation. On the right, the feedback form asks for name, registration number, email, and optional phone number. A subtitle further motivates students to share their experiences. Additional links like “Login” and navigation options make access convenient. The simple layout ensures a quick and user-friendly feedback submission process that helps strengthen system quality and user trust.

On the right side, the structured feedback form collects essential details such as the student’s name, registration number, email, and an optional phone number for further communication if needed. A helpful subtitle motivates students to provide honest and constructive feedback. The form also includes a text field where students can describe their experience or suggest improvements in detail, ensuring that the system receives meaningful input.

Convenient navigation links such as Login, Home, and other menu options increase accessibility, allowing students to move easily across the platform. The overall layout is simple, clean, and



user-friendly, reducing time and effort required submitting feedback. By enabling quick, easy, and thoughtful feedback submission.

SAMS

Home About Grievance Events Counseling Feedback Contact Login Sign Up

Home / Admin Login

Secure access to the administrative dashboard for managing the Student Affairs Management System.

**Admin Portal**

**Administrative Dashboard**

Manage students, events, grievances, and counseling applications with our comprehensive admin panel. Monitor system statistics and ensure smooth operations.

500+ Students

25+ Events

150+ Grievances

80+ Counseling

**Admin Access**

Please enter your admin credentials to continue.

Username \*

admin

Password \*

Enter admin password

Access Admin Panel

Need help? Contact Support

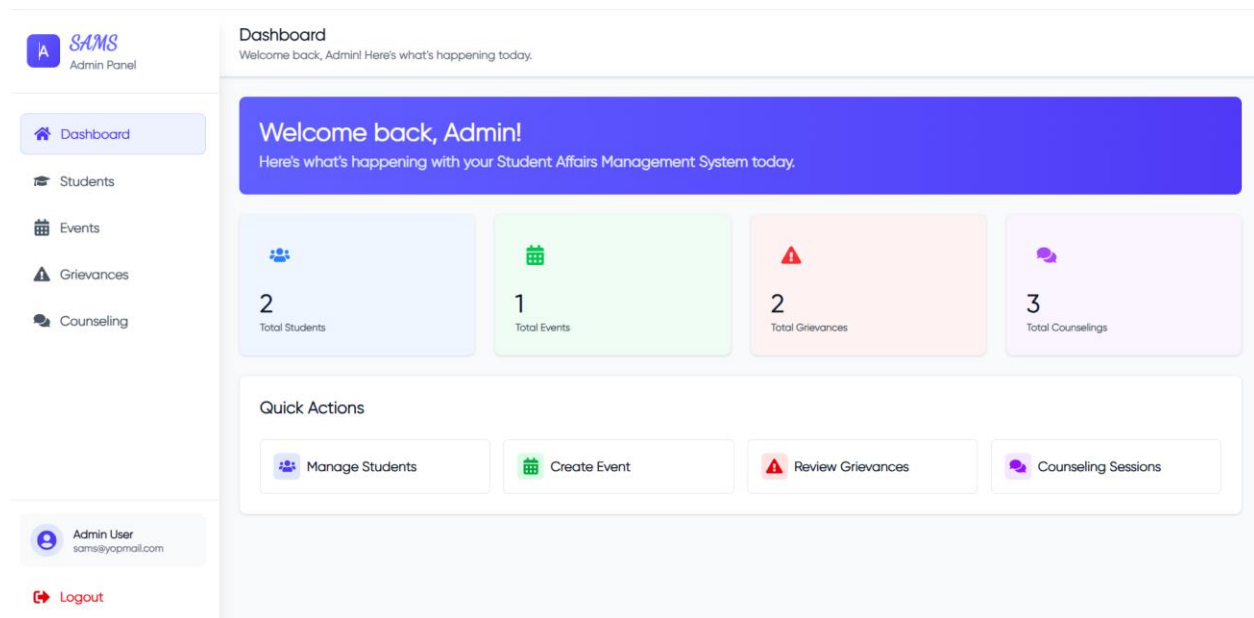
**Figure 5.10: Admin Login**

The Admin Login Page provides administrators with secure access to the SAMS management portal. On the left side, an introduction highlights the purpose of the portal, including a snapshot of system statistics such as the number of students, events, and submitted grievances. On the right, the login form allows administrators to enter their username and password to access the dashboard.

Additional links such as Need help provide support, while navigation bars enable access to basic sections of the system. The Access Admin Portal button ensures secure entry, maintaining the confidentiality of sensitive data. The page is designed with a professional and clean interface, emphasizing both security and ease of use. This login page is crucial for enabling administrators to manage student affairs, oversee system activities, and respond to submissions efficiently.

The Access Admin Portal button ensures secure entry, maintaining the confidentiality of sensitive data. The page is designed with a professional and clean interface, emphasizing both

security and ease of use. This login page is crucial for enabling administrators to manage student affairs, oversee system activities, and respond to submissions efficiently.



**Figure 5.11: Admin Dashboard**

The Admin Dashboard serves as the central hub for managing all aspects of student affairs within SAMS. A sidebar provides quick navigation to sections including Students, Events, Counseling, Grievances, and other key modules. At the top, a welcome message personalizes the experience for the logged-in admin. The dashboard displays summary cards showing total students, ongoing events, active counseling sessions, and submitted grievances, providing real-time insights into the system's operations. The Quick Actions section offers buttons for tasks such as managing students, creating events, and responding to grievances, streamlining administrative workflows. Admin user details, including name and email, are displayed for reference. The dashboard is organized for efficiency, allowing admins to quickly access necessary sections, monitor activities, and make informed decisions. Its design balances visual clarity with functional utility, ensuring that administrators can oversee and control SAMS operations.

# **Chapter 6**

## **Testing and Evaluation**

## **6 Testing and Evaluation**

This section covers the testing and evaluation processes performed on the SAMS platform to ensure all functionalities work as expected and meet the system requirements. It includes both manual and systematic approaches to verify modules such as Sign-Up, Login, Grievance Submission, Event Registration, and Counseling Booking.

### **6.1 Manual Testing**

Manual testing involves verifying each feature of SAMS by executing test cases without automation. The tester interacts with the system, inputs data, and observes outputs to confirm correct behavior. All user actions, such as signing up, logging in, submitting grievances, booking counseling sessions, and registering for events, are checked for accuracy and reliability.

#### **6.1.1 System Testing**

System testing in manual mode ensures that the SAMS platform functions as a whole. Testers verify end-to-end workflows, including student onboarding, dashboard access, grievance tracking, event registration, and counseling session management. Notifications, status updates, and reporting functionalities are also evaluated to confirm that the integrated system performs seamlessly under normal usage conditions.

#### **6.1.2 Unit Testing**

Unit testing focuses on testing individual components of the SAMS system, such as the Sign-Up form, Login module, Grievance submission form, Event module, and Counseling Booking form. Each module is tested separately to ensure it produces correct outputs for given inputs. For example, the Login module is tested for valid and invalid credentials, while the Grievance module is tested for proper ID assignment and status updates.[7]

##### **Unit Testing 1: Login as Student**

##### **Testing Objective:**

To verify that the student login system functions correctly, the testing process ensures that valid users are successfully authenticated and granted access to their accounts. At the same time, it

checks that invalid login attempts, such as incorrect registration numbers or passwords, are properly rejected.

**Table 6.1: Login as Student**

| Test Case ID | Test Case Name                         | Attributes & Values Tested                     | Expected Result                                     | Actual Result |
|--------------|----------------------------------------|------------------------------------------------|-----------------------------------------------------|---------------|
| UT-STU-001   | Student Login with valid credentials   | Username: SP22-BCS-151<br>Password: Password1! | Message: “User logged In Successfully!”             | Pass          |
| UT-STU-002   | Student Login with invalid credentials | Username: wrongUser<br>Password: wrongPass     | Message: “user does not exist”                      | Pass          |
| UT-STU-003   | Student login with empty fields        | Username:<br>Password:                         | Message: “registration number or password required” | Pass          |

## **Unit Testing 2: Logout as Student**

### **Testing Objective:**

To ensure the login and signup forms are functioning correctly for Students in SAMS website.

To ensure the login and signup forms on the SAMS website function correctly for students, the testing process involves verifying that all input fields accept valid data and reject invalid entries.

It checks that students can successfully create new accounts through the signup form and log in with the correct credentials.

Additionally, the system is evaluated for proper error handling, validation messages, and secure submission of user information, ensuring a smooth, reliable, and user-friendly experience for all students accessing the platform.

This testing also confirms that navigation between the forms and related pages works seamlessly without any issues.

**Table 6.2: Logout as Student**

| <b>Test Case ID</b> | <b>Test Case Name</b> | <b>Attributes &amp; Values Tested</b> | <b>Expected Result</b>               | <b>Actual Result</b> |
|---------------------|-----------------------|---------------------------------------|--------------------------------------|----------------------|
| UT-STU-004          | Student logout        | Click “Logout”                        | Message: “User logout successfully!” | Pass                 |

### **Unit Testing 3: Login as Admin**

#### **Testing Objective:**

To confirm that the admin login works accurately according to the given credentials, the testing process verifies that administrators with valid usernames and passwords can successfully access the SAMS management portal. It also ensures that invalid or incorrect credentials are properly rejected, and appropriate error messages are displayed.

Additionally, the system is checked for secure handling of sensitive information, smooth navigation to the dashboard after login, and reliable functionality under different test scenarios to maintain system integrity and security. . The testing further examines session management, including automatic logout and prevention of unauthorized access.

To confirm that the admin login works accurately according to the given credentials, the testing process verifies that administrators with valid usernames and passwords can successfully access the SAMS management portal. It also ensures that invalid or incorrect credentials are properly rejected, and appropriate error messages are displayed to guide the user. Additionally, the system is checked for secure handling of sensitive information, smooth navigation to the dashboard after

login, and reliable functionality under different test scenarios to maintain system integrity and overall security.

The testing further examines session management, including automatic logout, timeout handling, and the prevention of unauthorized access through restricted pages. This ensures that only authenticated administrators can manage system operations.

**Table 6.3: Login as Admin**

| <b>Test Case ID</b> | <b>Test Case Name</b>                | <b>Attributes &amp; Values Tested</b>       | <b>Expected Result</b>                         | <b>Actual Result</b> |
|---------------------|--------------------------------------|---------------------------------------------|------------------------------------------------|----------------------|
| UT-ADM-001          | Admin login with valid credentials   | Username: admin<br>Password: 00000000       | Message: “User logged In Successfully!”        | Pass                 |
| UT-ADM-002          | Admin login with invalid credentials | Username: wrongAdmin<br>Password: wrongPass | Message: “invalid user credentials”            | Pass                 |
| UT-ADM-003          | Admin login with empty fields        | Username:<br>Password:                      | Message: “Please fill out the username filed.” | Pass                 |

#### **Unit Testing 4: Logout as Admin**

##### **Testing Objective:**

To ensure that the admin can successfully log out from the system, the testing process verifies that clicking the logout button properly terminates the admin session and redirects the user to the login page. It also checks that all sensitive session data is cleared to prevent unauthorized access after logout.

Additionally, the system is evaluated to confirm that navigation to restricted pages is blocked once the admin has logged out, ensuring secure session management and maintaining the integrity and confidentiality of the SAMS platform.

**Table 6.4: Logout as Admin**

| Test Case ID | Test Case Name          | Attributes & Values Tested | Expected Result                     | Actual Result |
|--------------|-------------------------|----------------------------|-------------------------------------|---------------|
| UT-ADM-004   | Click the logout button | Click “Logout”             | Message: “Logged out successfully!” | Pass          |

### 6.1.3 Functional Testing

The functional testing will take place after the unit testing. In this functional testing, the functionality of each of the modules is tested. This is to ensure that the system produced meets the specifications and requirements.

#### Functional Testing 1: Student Module

##### Objective:

The objective of this functional testing is to verify that all core features of the Student Module in the SAMS system work accurately and reliably.

**Table 6.5: Login as Admin**

| Test Case ID | Test Case Name          | Attributes & Values Tested    | Expected Result                       | Actual Result |
|--------------|-------------------------|-------------------------------|---------------------------------------|---------------|
| FT-STU-001   | Student login (valid)   | sp22-bcs-151 / Password1!     | User logged In Successfully!          | Pass          |
| FT-STU-002   | Student login (invalid) | Wrong credentials             | “User does not exist”                 | Pass          |
| FT-STU-003   | View profile            | Open profile page             | Student profile displayed             | Pass          |
| FT-STU-004   | Edit profile            | Update name, password, number | Event successfully created and listed | Pass          |
| FT-STU-005   | Logout                  | Click logout                  | User logout successfully!             | Pass          |



The functional testing table for the Student Module outlines the key test cases designed to validate the core features available to students within the SAMS system. The first test case (FT-STU-001) verifies that a student with valid credentials such as sp22-bcs-151 / Password1 can successfully log into the system and access their dashboard. The second test case (FT-STU-002) checks the system's behavior when invalid login details are entered, ensuring that the system responds correctly by displaying the message User does not exist, which confirms that unauthorized access is restricted.

Next, the table includes functionality after login. Test case FT-STU-003 confirms that students can view their profile details by accessing the profile page, which should display all stored information accurately. Test case FT-STU-004 focuses on editing profile information, such as updating the student's name, password, or phone number. The expected result is that the changes are saved successfully and reflected immediately. Finally, test case FT-STU-005 verifies the logout functionality, making sure that when the student clicks the logout button, they are securely logged out of the system and returned to the login page.

Overall, the table ensures that all essential student-related features login, profile access, profile editing, and logout are functioning reliably, providing a smooth and secure user experience within SAMS.[8]

## **Functional Testing 2: Admin Module**

### **Objective:**

To ensure that after login, all the features available to society heads are fully functional, accessible, and responsive within the SAMS platform. This includes verifying that each module loads correctly, performs the intended actions, and handles user interactions smoothly. The objective also confirms that society heads can manage events, view student participation, and perform all assigned tasks without errors or interruptions.

All the features available to society heads are functional and responsive, the testing focuses on validating every module they interact with. This includes checking whether event creation, student management, grievance review, and announcements load correctly and respond without

delays. The goal is to confirm that the society head can smoothly navigate the dashboard, perform all assigned tasks, and receive accurate feedback from the system. By thoroughly validating each feature, the system ensures a seamless workflow, reduces errors, and supports efficient society operations.

**Table 6.6: Login as Admin**

| <b>Test Case ID</b> | <b>Test Case Name</b> | <b>Attributes &amp; Values Tested</b> | <b>Expected Result</b>       | <b>Actual Result</b> |
|---------------------|-----------------------|---------------------------------------|------------------------------|----------------------|
| FT-ADM-001          | Admin login (valid)   | admin / 00000000                      | User logged In Successfully! | Pass                 |
| FT-ADM-002          | Admin login (invalid) | Wrong credentials                     | Invalid user credentials     | Pass                 |
| FT-ADM-003          | Logout                | Click logout                          | Logged out successfully!     | Pass                 |

This table describes the functional testing performed on the Admin Module of the SAMS system to ensure that key administrative actions such as logging in and logging out work reliably. Each test case verifies a specific behavior to confirm that the system responds correctly under both valid and invalid conditions.

The first test case (FT-ADM-001) checks whether the admin can successfully log in using valid credentials. When the correct username and password (admin / 00000000) are entered, the system should authenticate the user and redirect them to the admin dashboard. The expected result matches the actual outcome, confirming the login process works as intended.

The second test case (FT-ADM-002) evaluates how the system handles invalid login attempts. When incorrect credentials are entered, the system is expected to display an appropriate error message such as Invalid user credentials. The Pass status indicates that the system correctly prevented unauthorized access.

The third test case (FT-ADM-003) verifies the logout functionality. When the logout button is clicked, the system should terminate the admin session and redirect the user to the login page.

The successful match between expected and actual result confirms that session handling and logout operations are functioning properly.[9]

### 6.1.4 Integration Testing

Integration testing is a crucial phase in software development that focuses on combining individual units or modules and testing them as a group. This process helps identify defects in the interactions between modules, ensuring they communicate correctly, and function together as intended. By detecting interface issues, logical errors, and potential data mismatches, integration testing confirms that the application operates smoothly in a real-world environment. It ultimately enhances software reliability before proceeding to further testing stages.

#### Objective:

To ensure that the admin panel integrates correctly with various modules like user, society, and event management.[10]

**Table 6.7: Login as Student**

| Test Case ID | Test Case Name                 | Integrated Modules | Expected Result                      | Actual Result |
|--------------|--------------------------------|--------------------|--------------------------------------|---------------|
| IT-STU-001   | Student login & load dashboard | Login + Dashboard  | Student can submit forms after login | Pass          |
| IT-STU-002   | Student updates profile        | Profile + Database | Updates stored successfully          | Pass          |
| IT-STU-003   | Student logout                 | Dashboard + Logout | Logout successfully                  | Pass          |

The integration testing of the SAMS system focuses on verifying that multiple modules work together seamlessly and that data flows correctly between them. Test Case IT-STU-001 checks the integration between the Student Login module and the Dashboard. It ensures that after a successful login, the student can access the dashboard and perform actions such as submitting forms. The expected outcome is smooth access and functionality, and the actual result confirmed that the student could submit forms successfully, resulting in a Pass.[11]

Test Case IT-STU-002 examines the integration between the Profile module and the Database. It verifies that when a student updates their profile, the changes are correctly stored in the database. The test confirms that updates such as changes to the name, password, or contact details are reflected accurately, and the actual result matched expectations, leading to a Pass.

Test Case IT-STU-003 tests the interaction between the Dashboard and Logout modules. It ensures that when a student logs out from the dashboard, the session terminates properly, and the system securely redirects the user to the login page. The actual result confirmed successful logout functionality, yielding a Pass.

## Integration Testing 2: Admin Module

**Objective:** To ensure that the correct screen and modules load for society president, and all features work as expected.

**Table 6.8: Login as Admin**

| Test Case ID | Test Case Name             | Attributes & Values Tested | Expected Result         | Actual Result |
|--------------|----------------------------|----------------------------|-------------------------|---------------|
| IT-ADM-001   | Admin login & access panel | Login + Admin Panel        | Admin panel opens       | Pass          |
| IT-ADM-002   | Admin manages users        | User Management + Database | Status updated          | Pass          |
| IT-ADM-003   | Admin logout               | Admin Panel + Logout       | Logged out successfully | Pass          |

The integration testing for the Admin Module ensures that the system's administrative functions work cohesively and all connected modules operate correctly. Test Case IT-ADM-001 tests the integration between the Admin Login module and the Admin Panel. It verifies that upon entering valid credentials, the admin panel opens successfully, allowing administrators to access management features such as student monitoring, event management, and grievance tracking. The actual result confirmed that the admin panel opened as expected, resulting in a Pass. Test Case IT-ADM-002 focuses on the integration between the User Management module and the

Database. This test ensures that when an administrator updates user details or status, the changes are correctly reflected in the database. The expected outcome is accurate status updates, and the actual result confirmed proper data handling, leading to a Pass. Test Case IT-ADM-003 examines the interaction between the Admin Panel and the Logout module.[12]

## 6.1 Tools and Technologies

| <b>Tools And Technologies</b> | <b>Tools</b>                                            | <b>Version</b>                              | <b>Rationale</b>     |
|-------------------------------|---------------------------------------------------------|---------------------------------------------|----------------------|
|                               | MS Visual Studio                                        | 2023                                        | Web based project    |
|                               | MongoDB/Postgresql                                      | 6.0                                         | DBMS                 |
|                               | MS Project                                              | 2021                                        | Gantt Chart          |
|                               | Figma                                                   |                                             | Mockups              |
|                               | Adobe Photoshop                                         | CSC 6                                       | Design Work          |
|                               | MS Word                                                 | 2021                                        | Documentation        |
|                               | MS Power Point                                          | 2021                                        | Presentation         |
|                               | <b>Technology</b>                                       | <b>Version</b>                              | <b>Rationale</b>     |
|                               | JavaScript/Typescript                                   | ES6                                         | Programming language |
|                               | Web                                                     | Latest                                      | Platform             |
|                               | Postgresql                                              | Latest                                      | Query Language       |
|                               | Mern Stack<br>(React.js/Next.js<br>+<br>Nodejs(Nestjs)) | Next.js 15<br>React.js 19<br>Node.js 23.3.0 | Admin                |

## 6.2 Summary

This chapter discussed various types of testing performed on the SAMS platform, including manual testing, functional testing, and integration testing. Each module, such as Student Login, Sign-Up, Profile Management, Grievance Submission, Event Registration, Counseling Booking,

and Admin functionalities, was tested using both valid and invalid inputs. The test cases covered typical user scenarios as well as edge cases to ensure robustness.

The system's responses were compared against expected outcomes, and the final status of each test case was recorded as Pass or Fail. The testing confirmed that all modules function correctly, interact seamlessly, and maintain data integrity while providing a secure and user-friendly experience.[13]

# **Chapter 7**

## **Conclusion and Future Work**

## **7 Conclusion and Future Work**

### **7.1 Conclusion**

SAMS has been designed to streamline student services, simplify event participation, and enhance communication between students, counselors, and administrators. The main goal of this project was to create a digital platform where students can register for the system, submit grievances, book counseling sessions, stay updated on university events, and access all related services in one place. For administrators, SAMS provides tools to manage student records, approve and track grievances, monitor events, and oversee the overall system efficiently.

SAMS has successfully made the management of student affairs more organized, user-friendly, and accessible. Traditionally, students had to rely on manual forms, physical notice boards, or informal communication to submit grievances, register for events, or request counseling. Now, with SAMS, all these processes are digital and centralized, saving both time and effort for students and administrators alike.

Another major advantage of SAMS is its clean and intuitive interface. Users do not need any specialized training to navigate the platform. Whether a student is submitting a grievance, registering for an event, booking a counseling session, or checking updates, or whether an admin is managing records and monitoring system activities, the platform offers smooth and simple navigation for all users.

From development to testing, the quality of SAMS has been ensured by thoroughly checking all major modules, including login, sign-up, profile management, grievance submission, event registration, counseling booking, feedback collection, and administrative controls. All modules have been tested for both valid and invalid scenarios, and they have been found to work as expected.[14]

In conclusion, SAMS has fulfilled its purpose of improving student engagement, simplifying administrative tasks, and providing a centralized, secure, and efficient system for managing



student affairs. It represents a significant step toward making campus services more organized, technology-driven, and accessible for both students and administrators.

## **7.2 Future Work**

Even though the current version of the SAMS works well and meets its objectives, there is always room for improvement and expansion. Here are some features and ideas we plan to add in the future to make the app even better:

### **7.2.1 Automated Notification**

Enhancing SAMS to send automated alerts via email or SMS for events, grievance status updates, and counseling appointments to improve communication and ensure students never miss important updates.

### **7.2.2 Mobile Application Integration**

Developing a mobile app version of SAMS to provide students with on-the-go access to their dashboard, event registrations, counseling bookings, and notifications, making the system more accessible and user-friendly.

### **7.2.3 Advanced Analytics and reporting for Admins**

Implementing advanced reporting and analytics tools to allow administrators to monitor student participation, grievance trends, and event effectiveness, enabling data-driven decision-making.

### **7.2.4 Multi-Language Support**

Add support for different languages (like Urdu) so that more users can understand and use the app easily.[15]

### **7.2.5 Strong Security Measure**

Improve data protection by adding advanced login features like two-factor authentication (2FA) and better encryption.

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